ISWE406P

Agile Development Process and DevOps Lab

Assessment-2

Faculty: Dr. Sweta Bhattacharya
Slot: L3+L4

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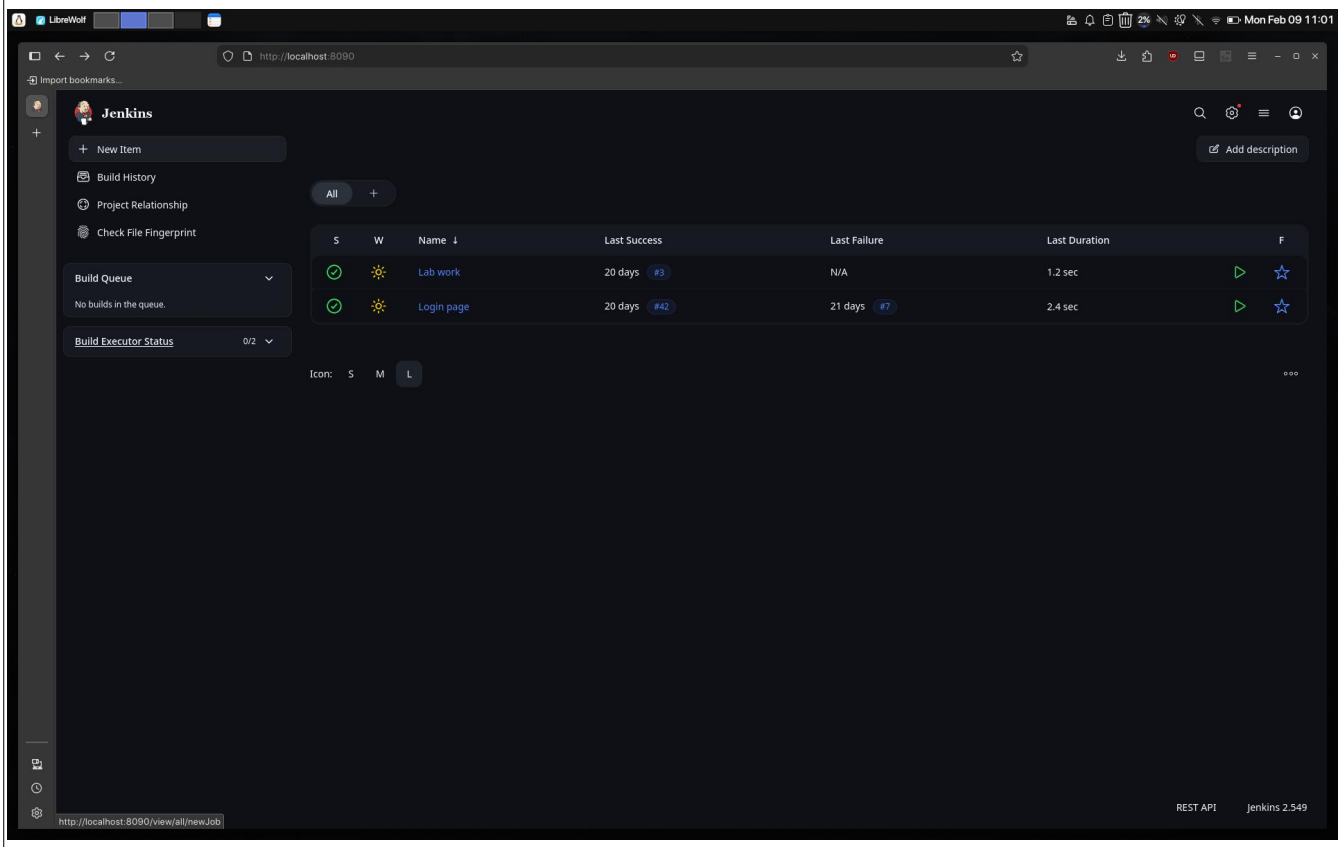
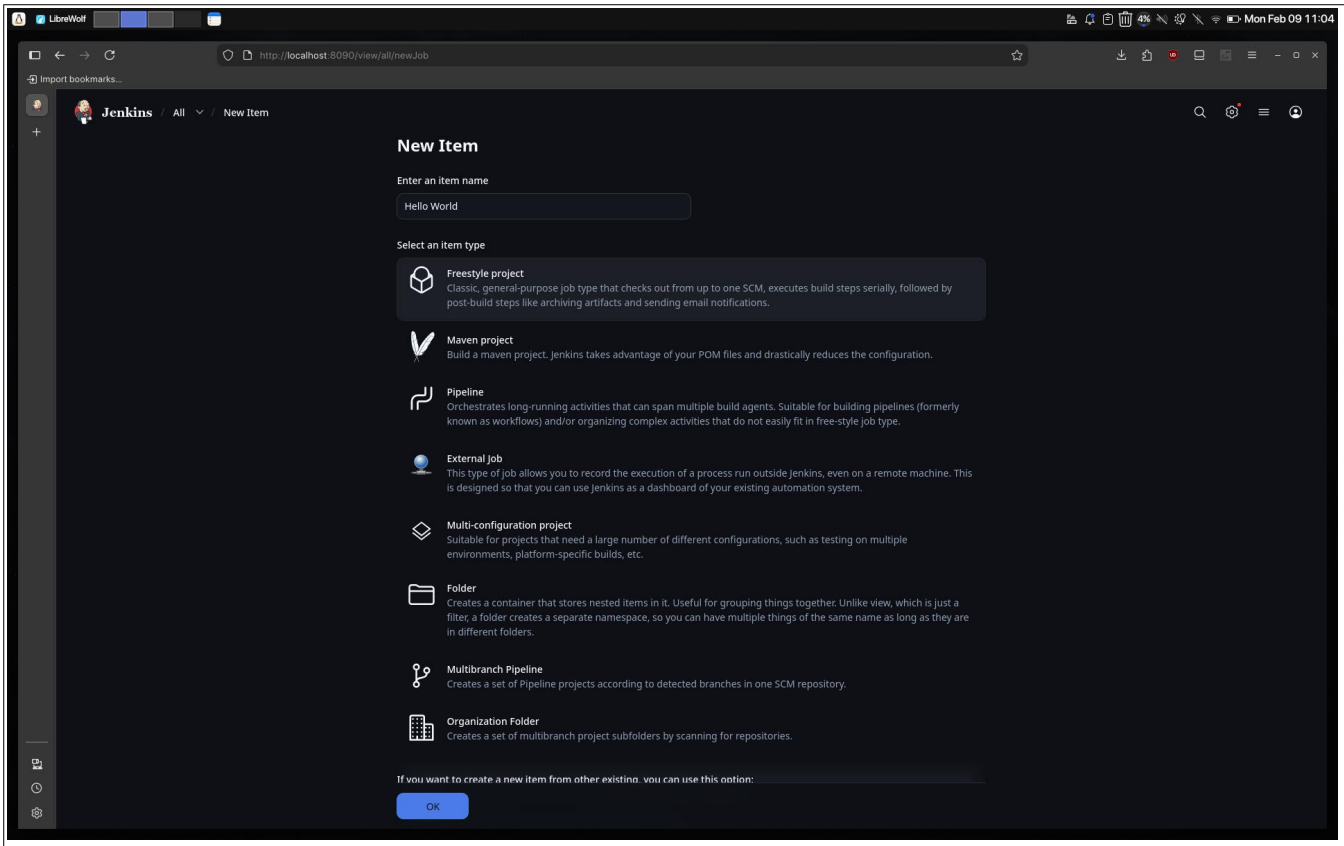
Reg No: 23MIS0170

1. Understand what Jenkins is and run a job successfully.

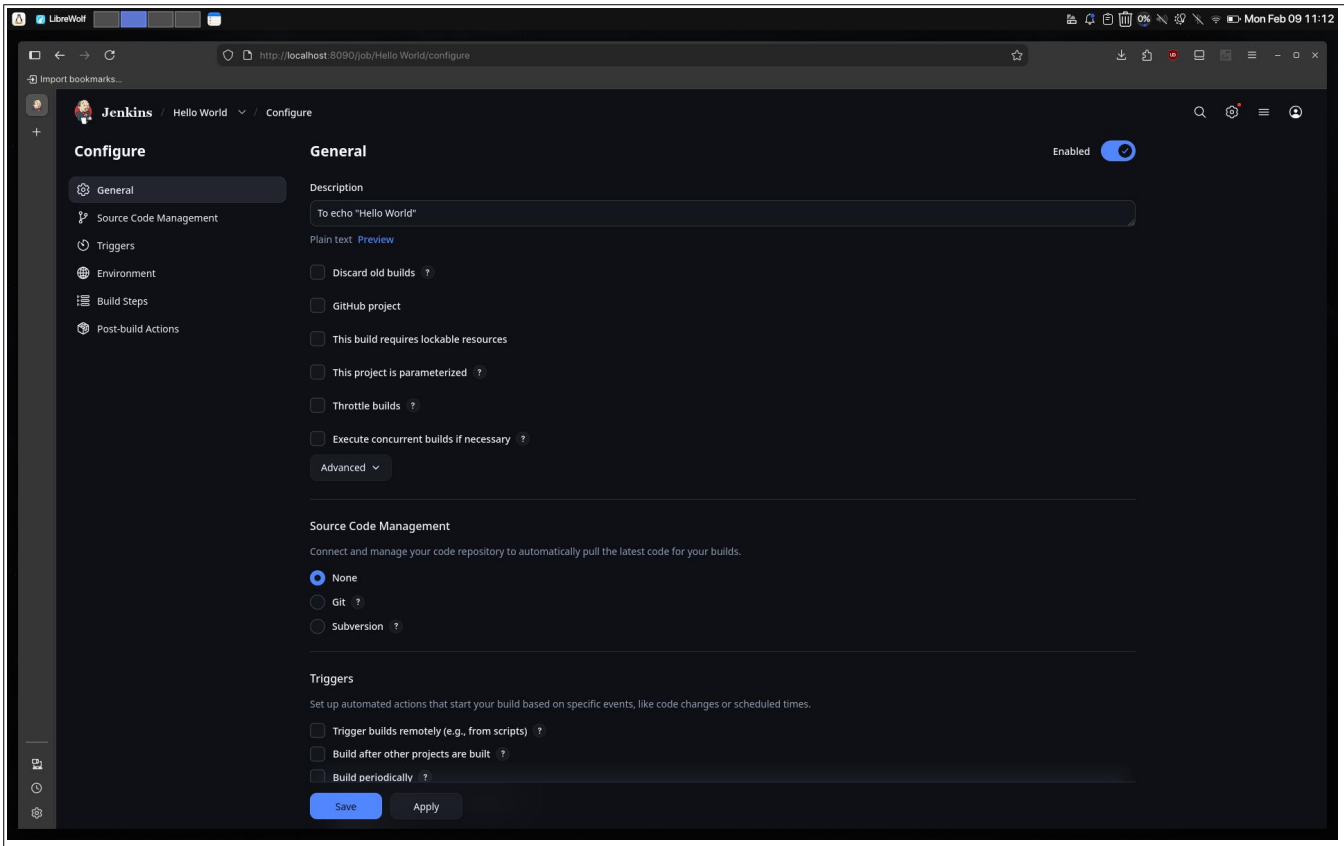
What is CI/CD?
CI/CD (Continuous Integration and Continuous Delivery/Deployment) is a software development practice where code changes are automatically built, tested, and delivered through a pipeline. It helps teams release software faster, with fewer errors and more reliability.

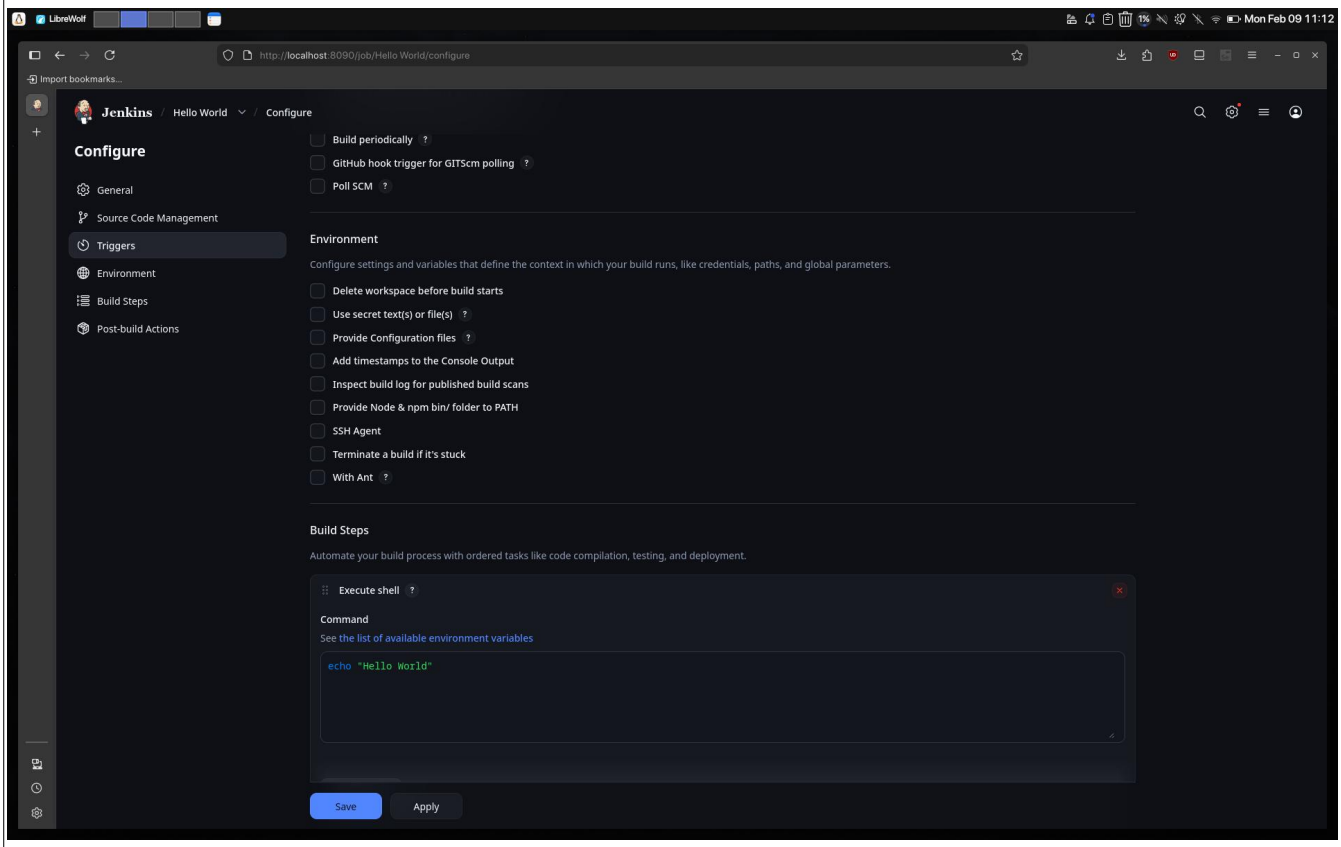


click **new item**

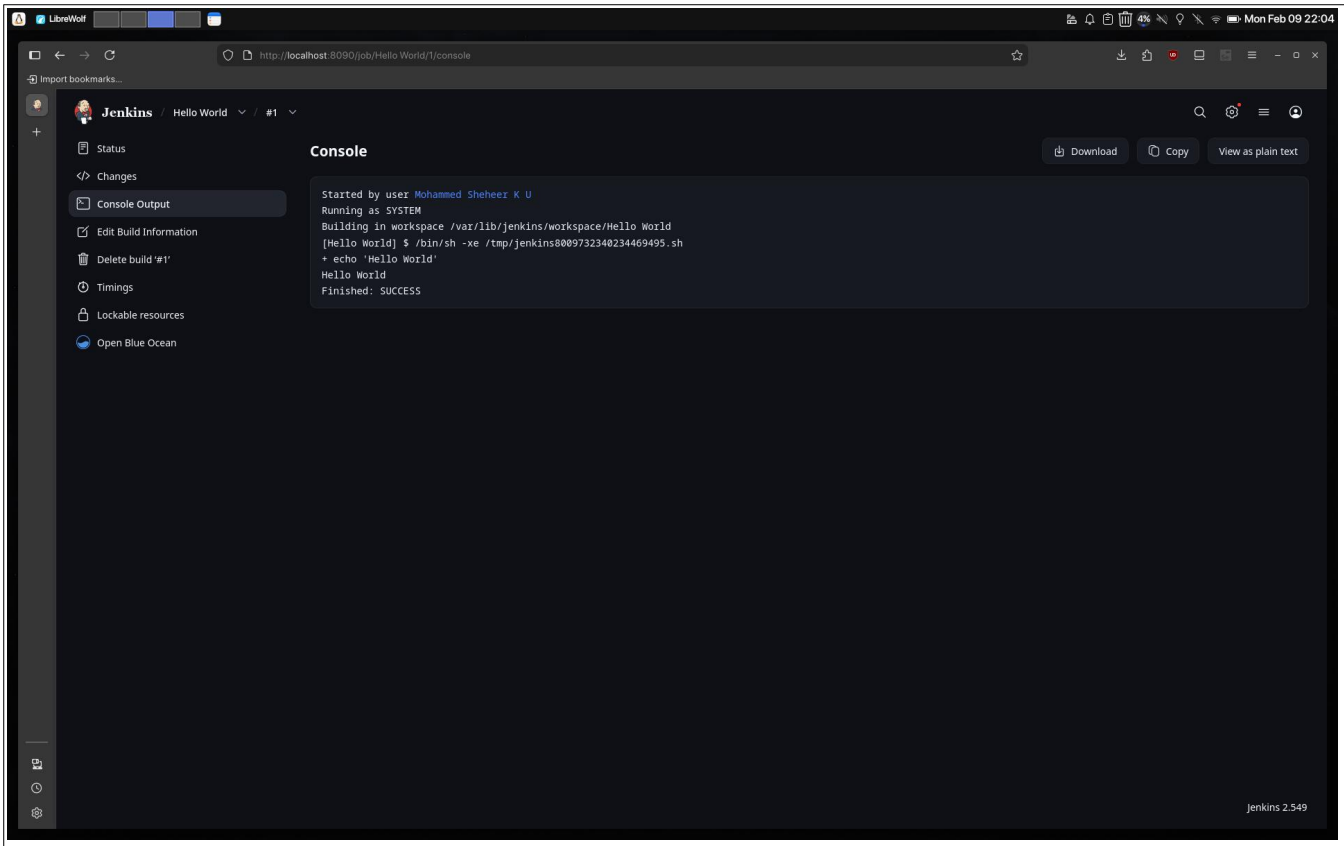


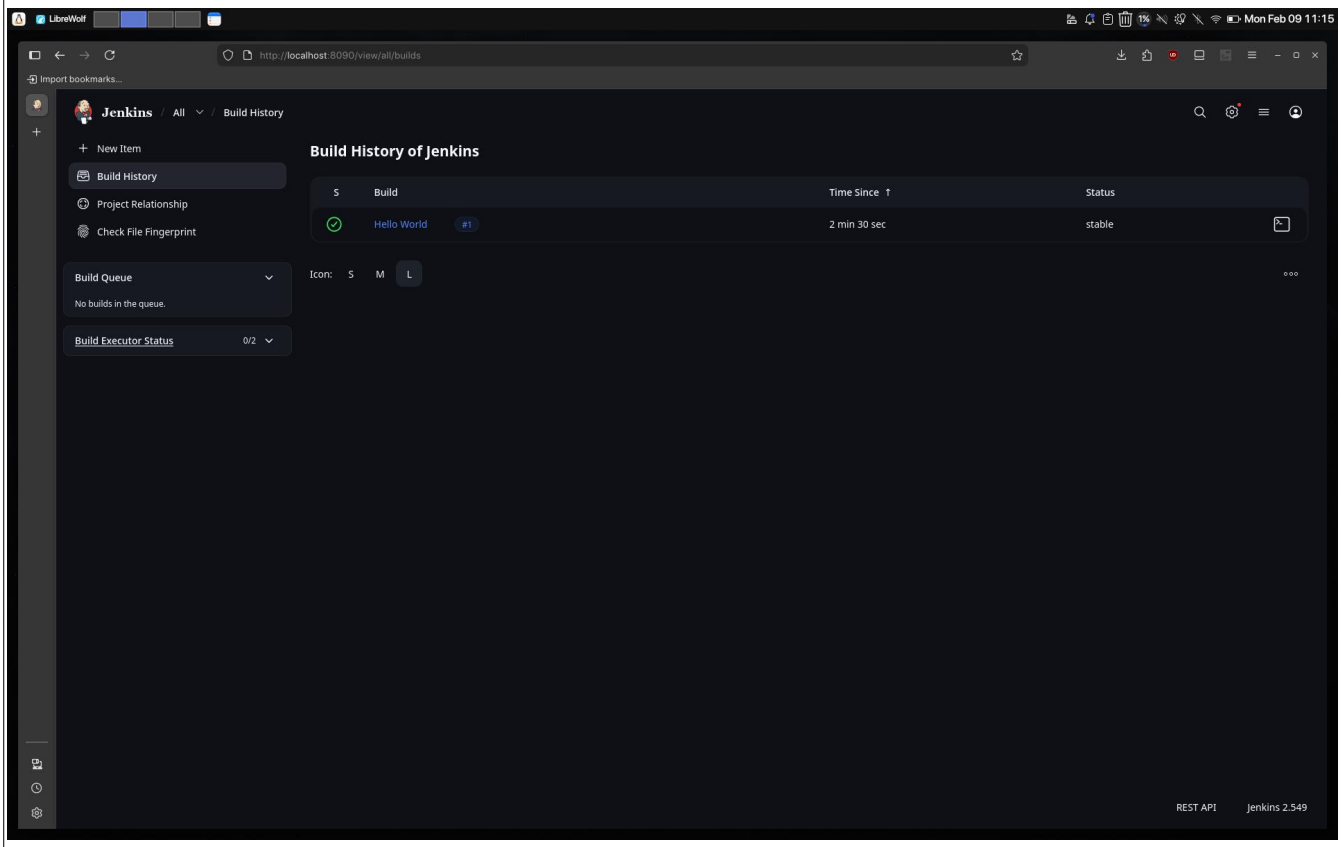
create a new freestyle project





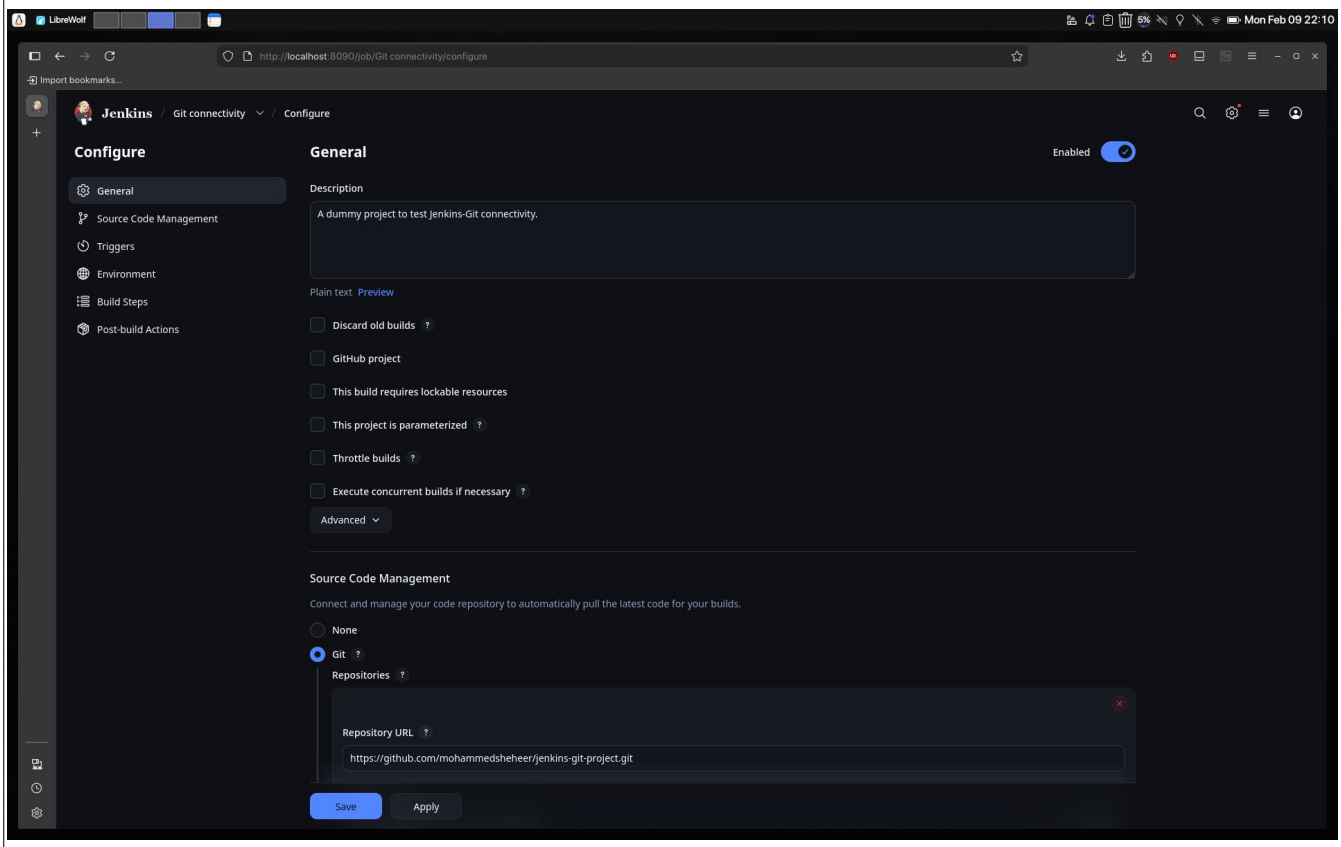
click **Save** and **Apply**

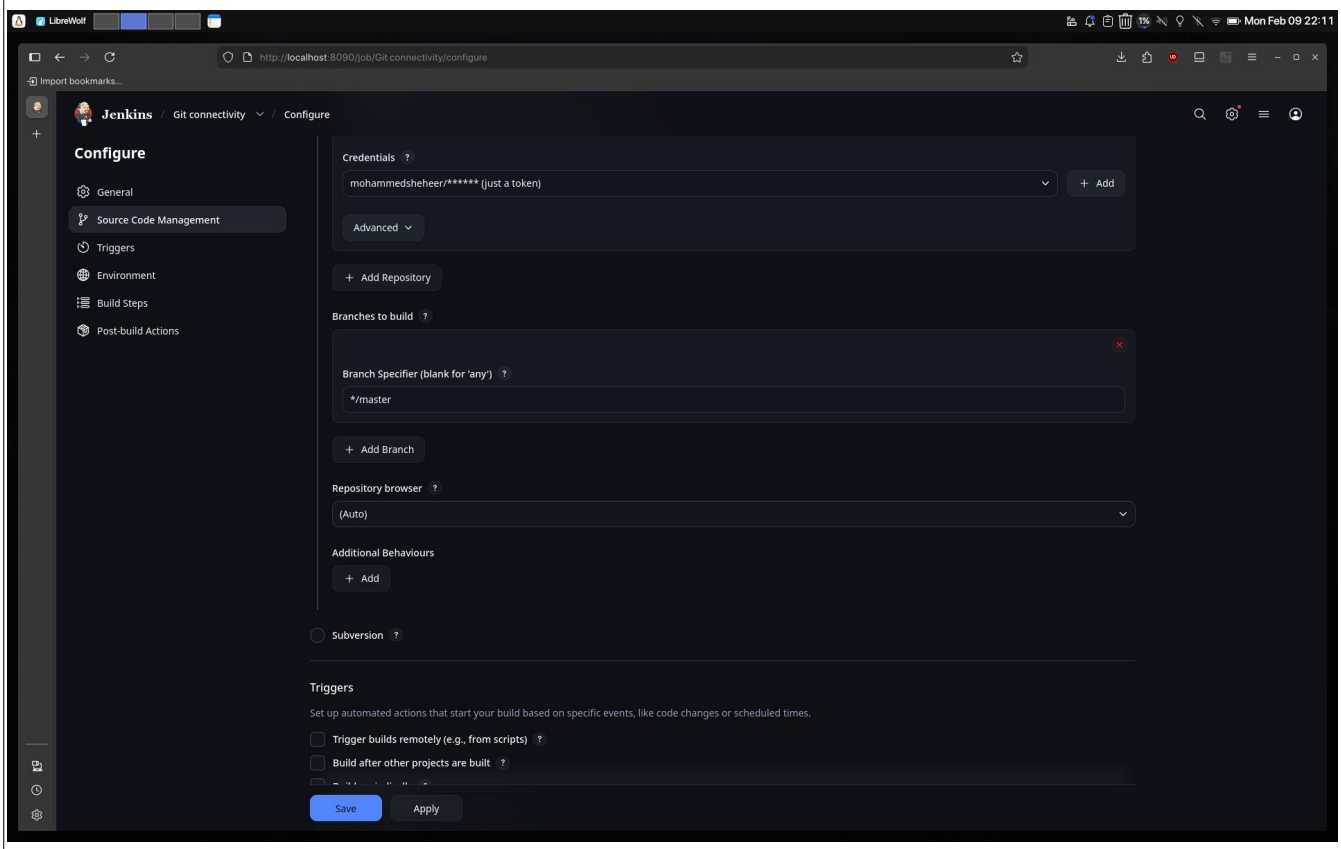


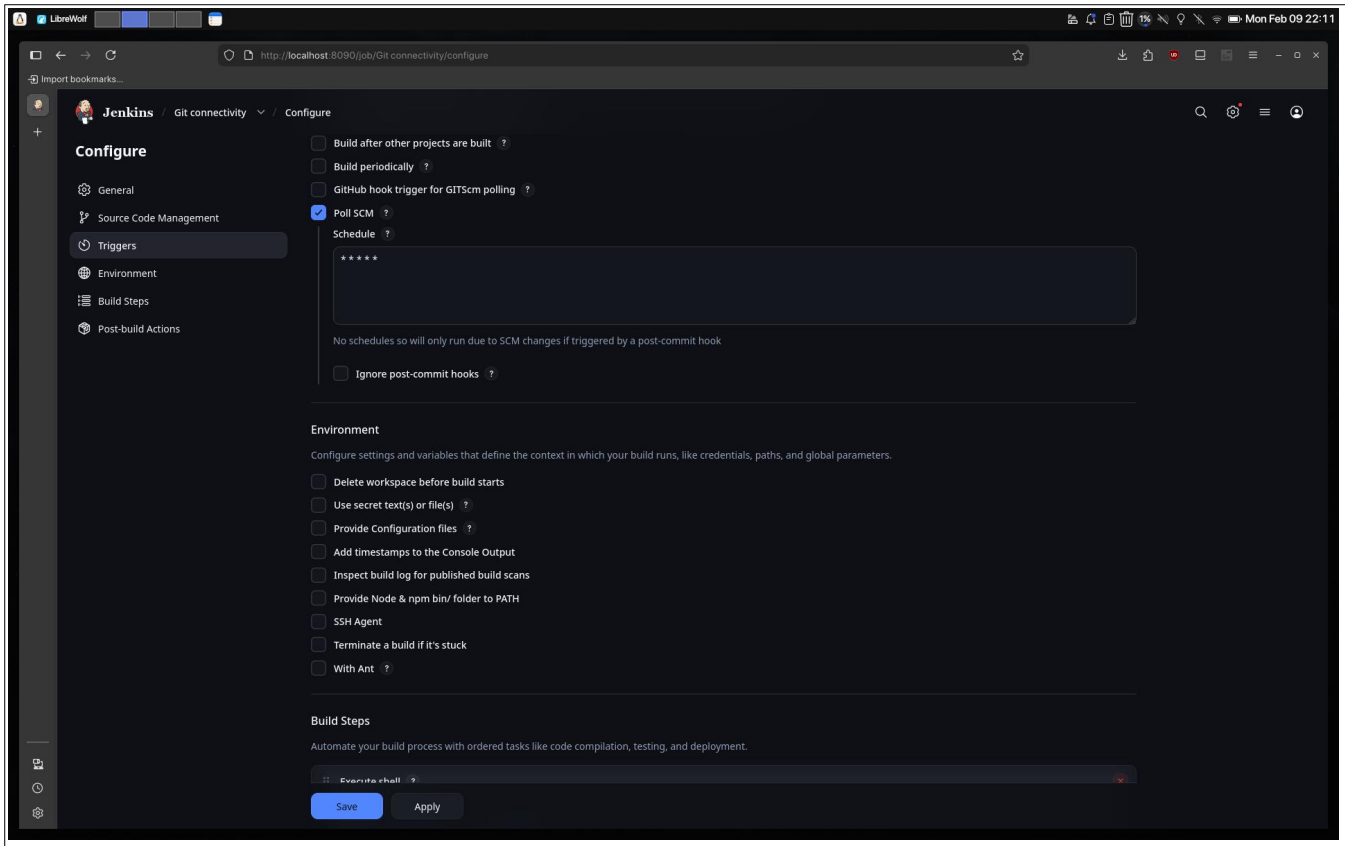


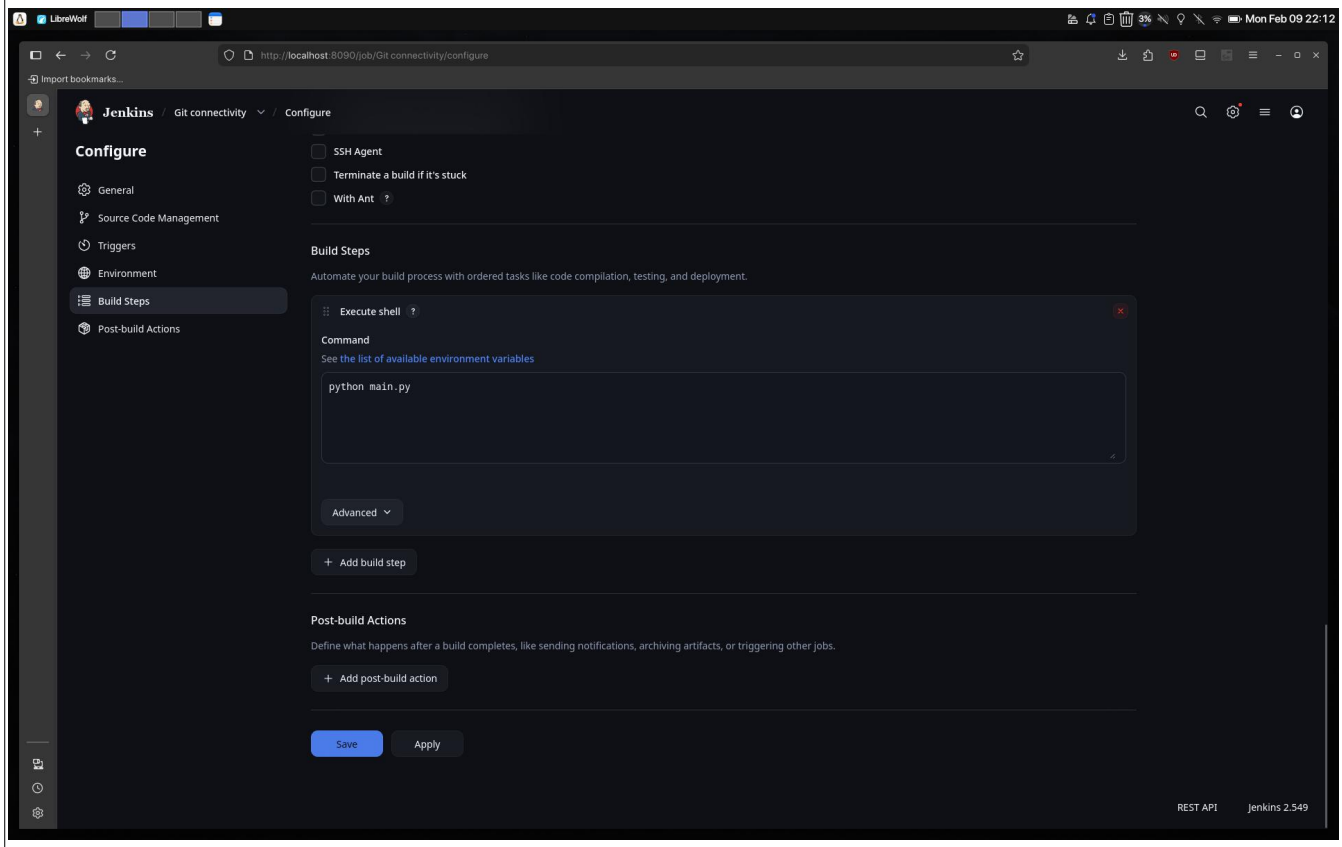
build history

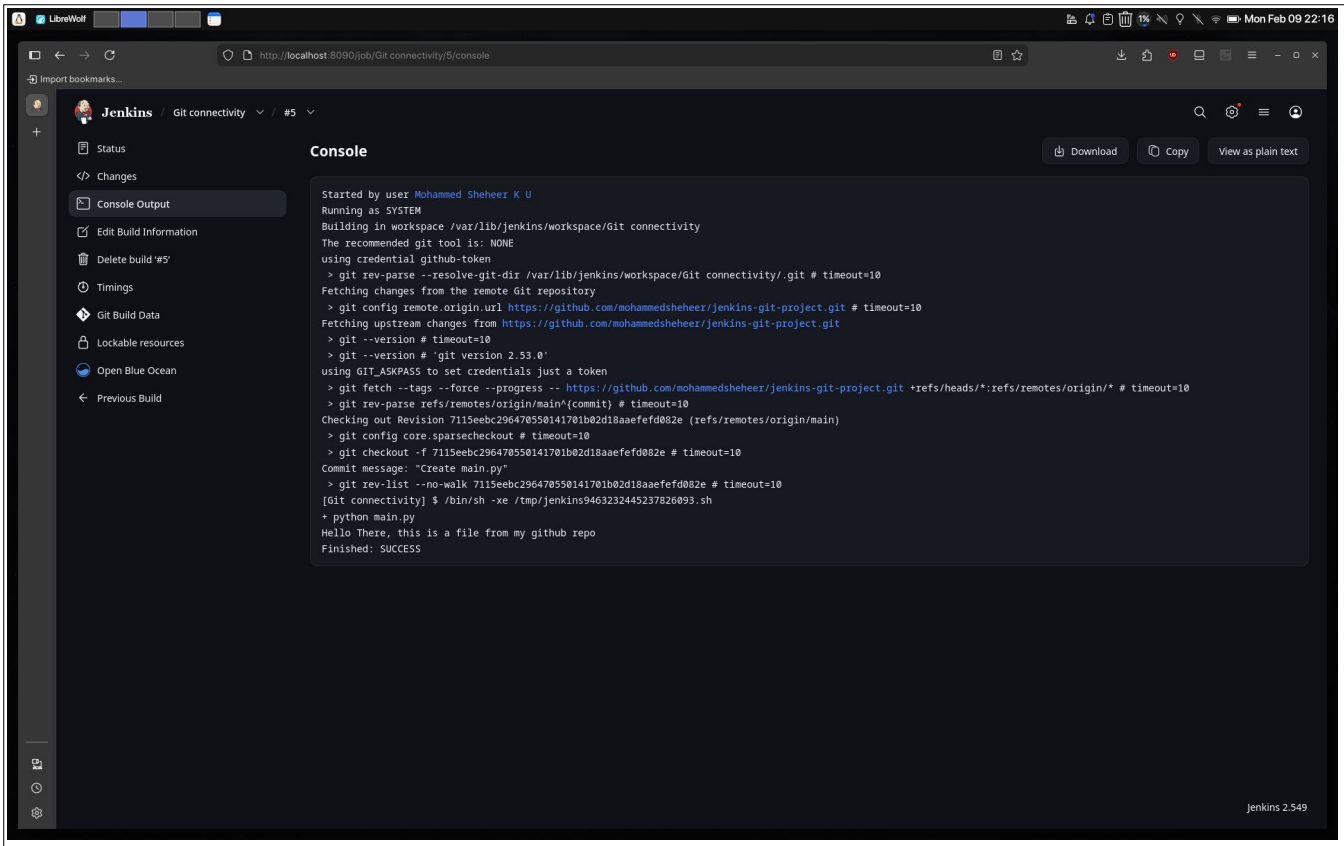
2. Work with source code and build automation.



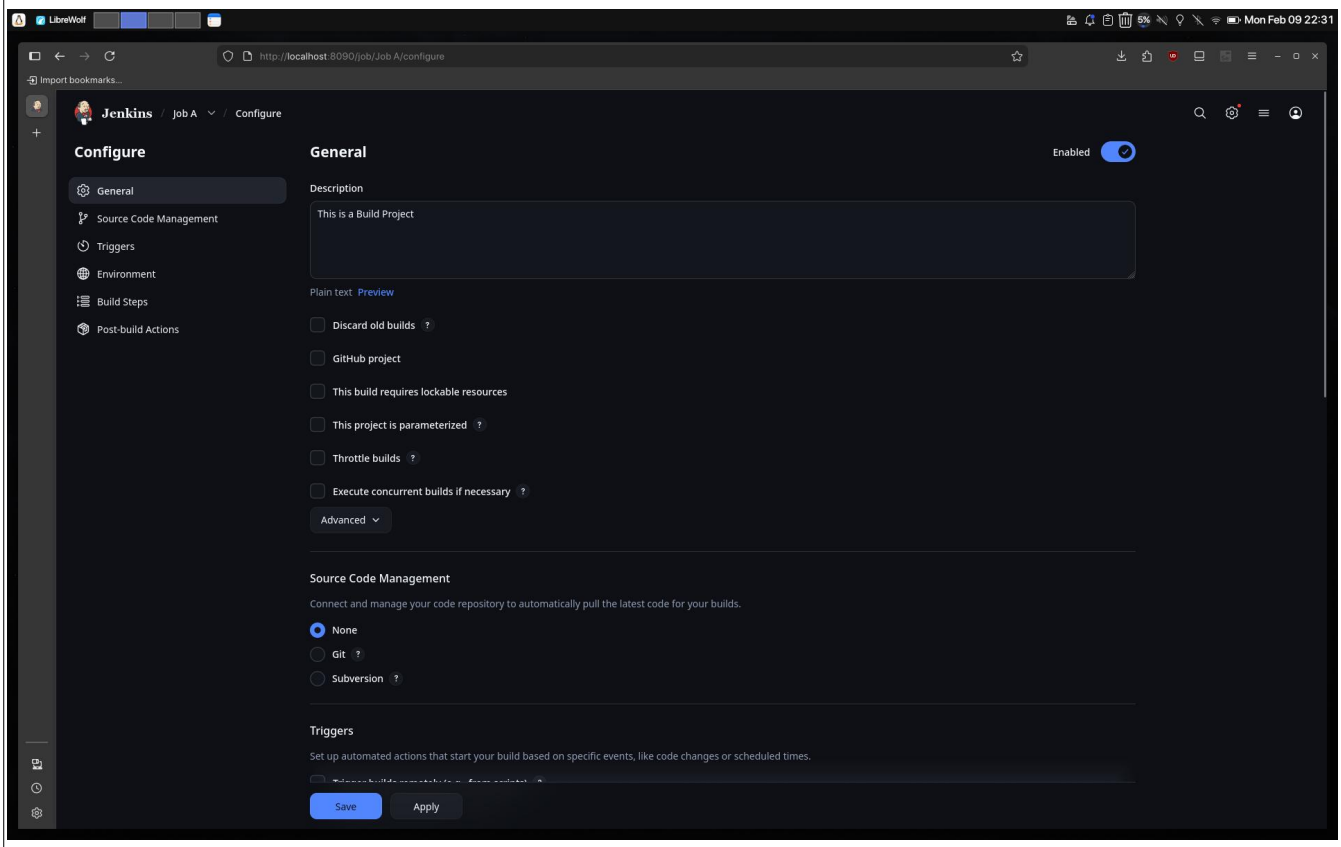


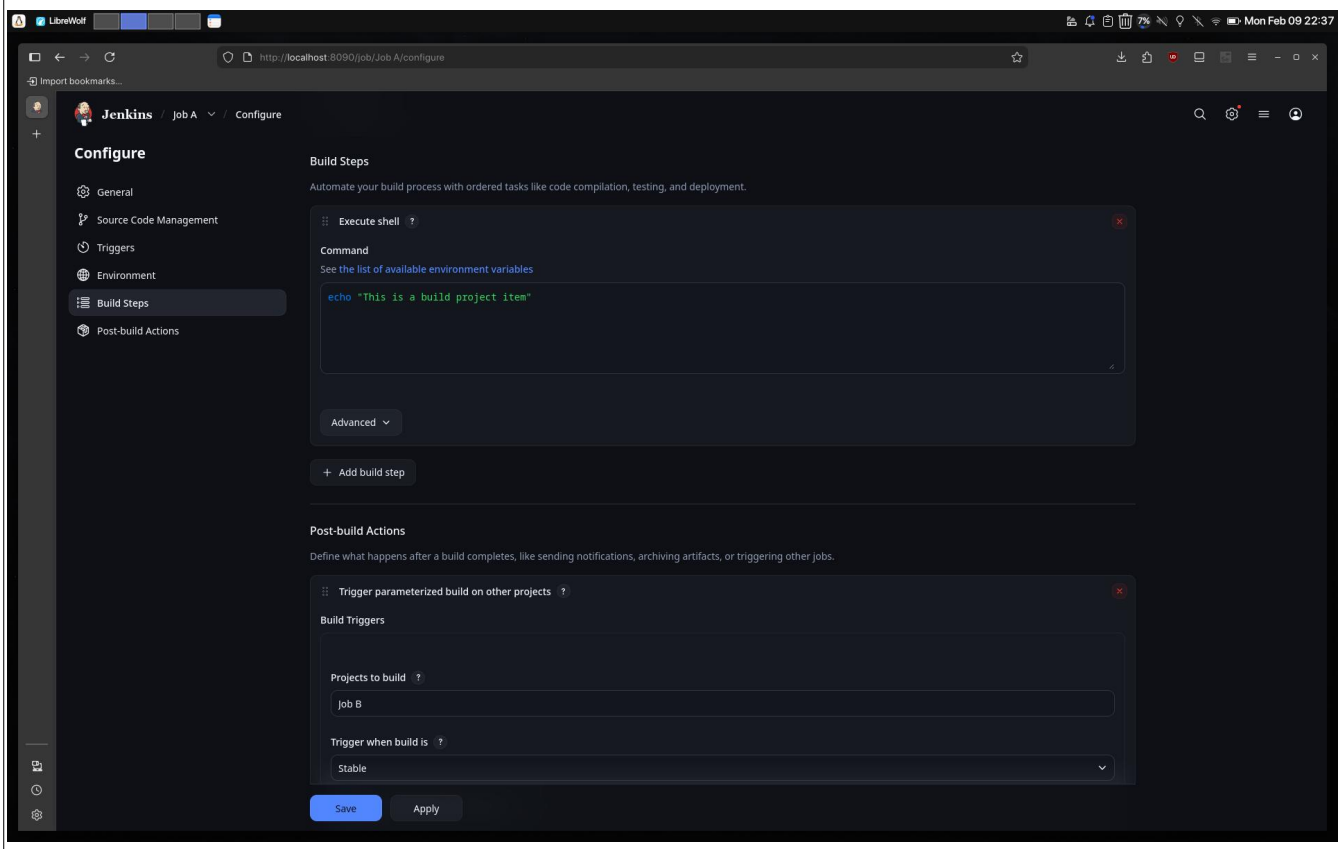

LibreWolf
Mon Feb 09 22:11
http://localhost:8090/job/Git connectivity/configure
Import bookmarks...
Jenkins / Git connectivity / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build after other projects are built
Build periodically
GitHub hook trigger for GITScm polling
Poll SCM
Schedule
* * * * *
No schedules so will only run due to SCM changes if triggered by a post-commit hook
Ignore post-commit hooks
Environment
Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.
Delete workspace before build starts
Use secret text(s) or file(s)
Provide Configuration files
Add timestamps to the Console Output
Inspect build log for published build scans
Provide Node & npm bin/ folder to PATH
SSH Agent
Terminate a build if it's stuck
With Ant
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Save
Apply

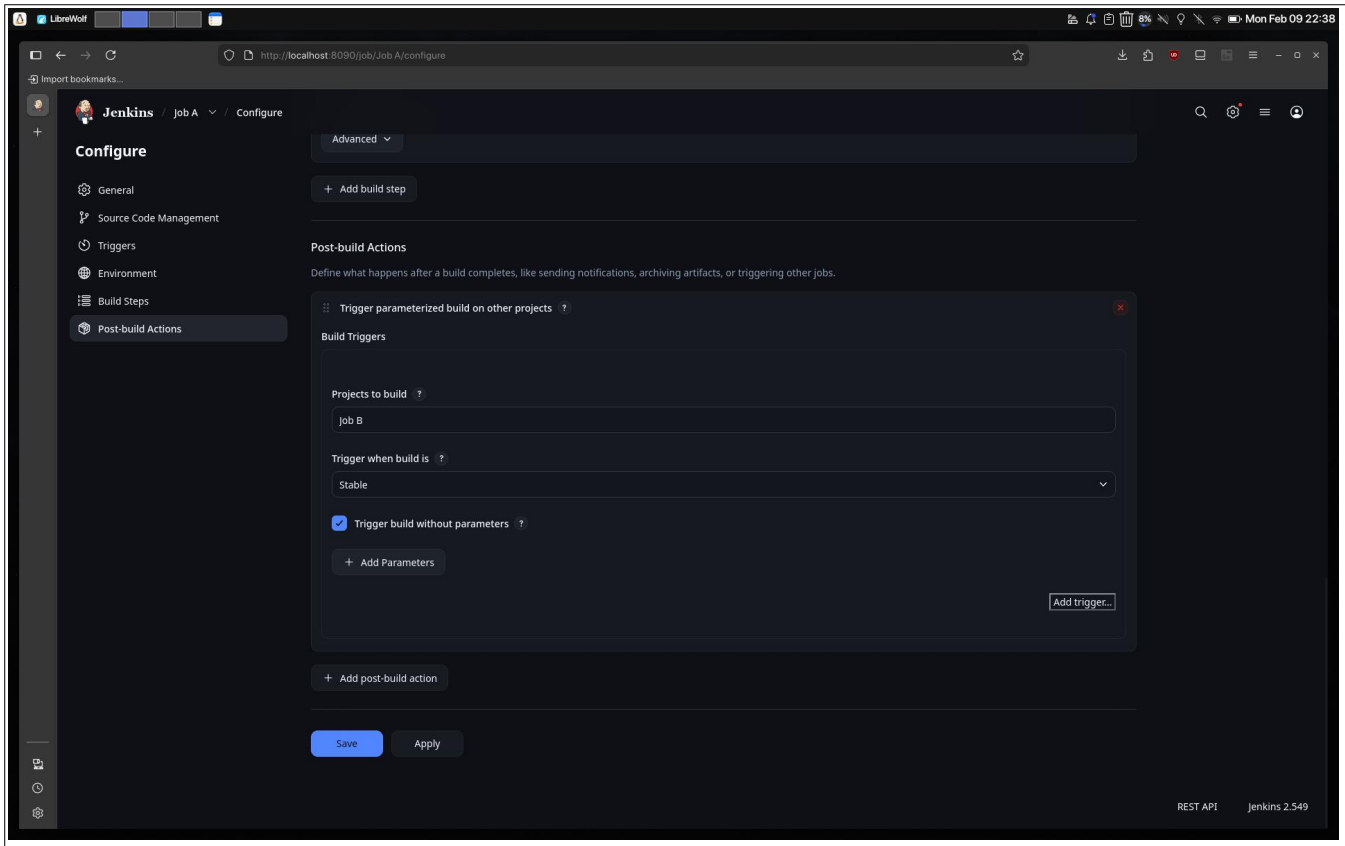

LibreWolf
Mon Feb 09 22:12
http://localhost:8090/job/Git connectivity/configure
Import bookmarks...
Jenkins / Git connectivity / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
SSH Agent
Terminate a build if it's stuck
With Ant
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute shell
Command
See the list of available environment variables
python main.py
Advanced
Add build step
Post-build Actions
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
Add post-build action
Save
Apply
REST API
Jenkins 2.549

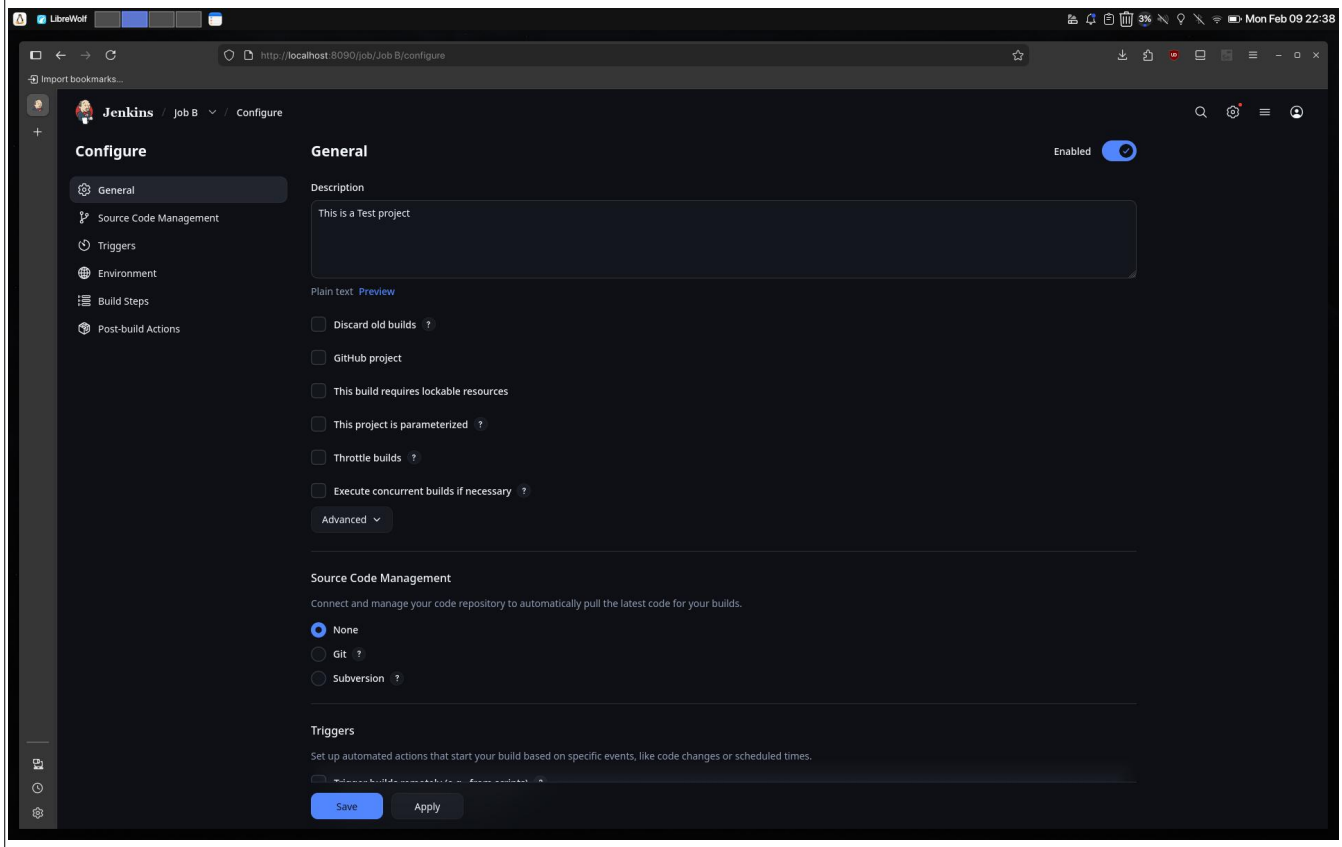

LibreWolf
Mon Feb 09 22:16
http://localhost:8090/job/Git connectivity/5/console
Import bookmarks...
Jenkins / Git connectivity / #5
Status
Changes
Console Output
Edit Build Information
Delete build '#5'
Timings
Git Build Data
Lockable resources
Open Blue Ocean
Previous Build
Console
Download
Copy
View as plain text
Started by user Mohammed Sheheer K U
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Git connectivity
The recommended git tool is: NONE
using credential github-token
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Git connectivity/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/mohammedsheheer/jenkins-git-project.git # timeout=10
Fetching upstream changes from https://github.com/mohammedsheheer/jenkins-git-project.git
> git --version # timeout=10
> git --version # 'git version 2.53.0'
using GIT_ASKPASS to set credentials just a token
> git fetch --tags --force --progress -- https://github.com/mohammedsheheer/jenkins-git-project.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 7115eebc296470550141701b02d18aaefefd082e (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 7115eebc296470550141701b02d18aaefefd082e # timeout=10
Commit message: "Create main.py"
> git rev-list --no-walk 7115eebc296470550141701b02d18aaefefd082e # timeout=10
[Git connectivity] $ /bin/sh -xe /tmp/jenkins9463232445237826093.sh
+ python main.py
Hello There, this is a file from my github repo
Finished: SUCCESS
Jenkins 2.549

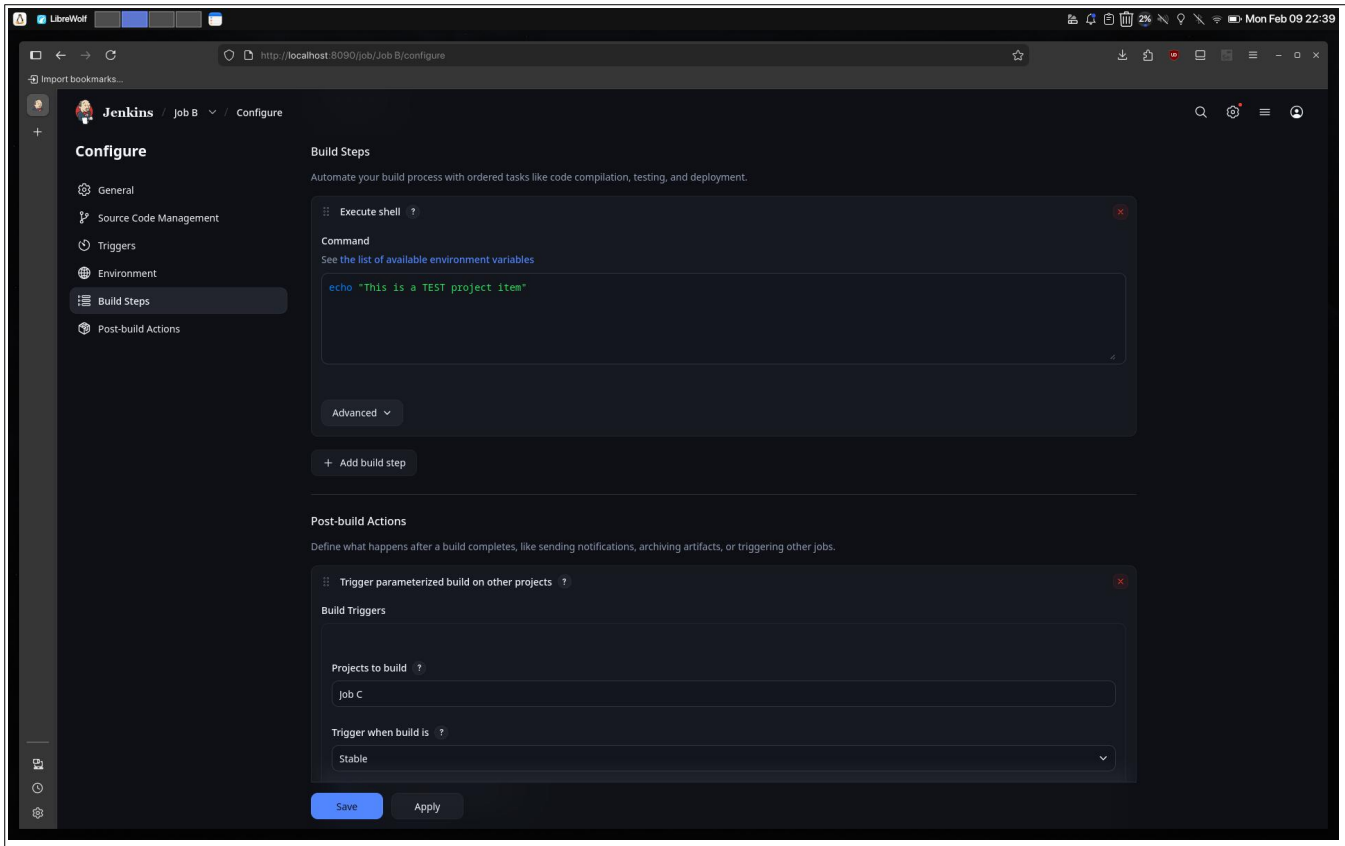
3. Understand job flow visually.

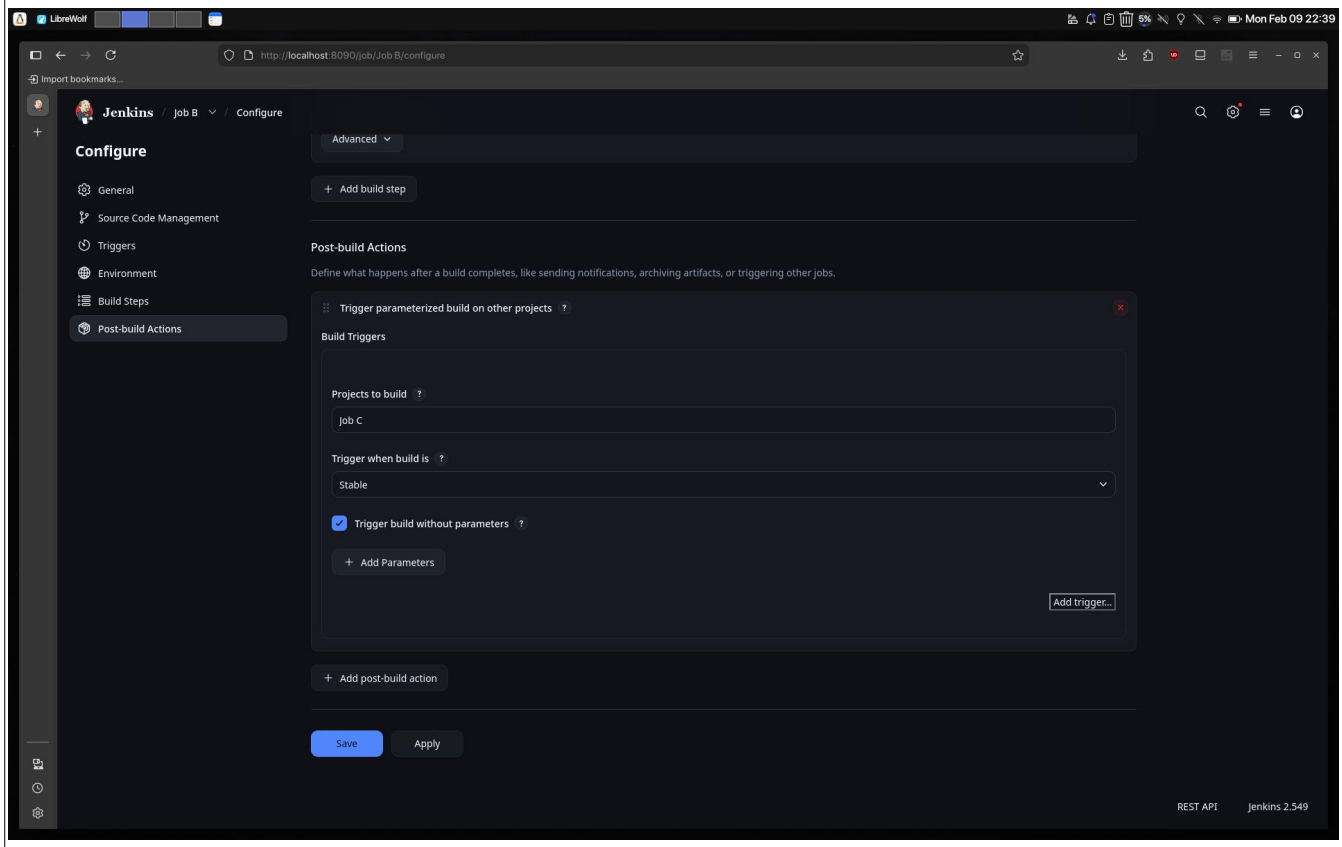





LibreWolf
Mon Feb 09 22:38
http://localhost:8090/job/Job A/configure
Import bookmarks...
Jenkins / Job A / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Advanced
Add build step
Post-build Actions
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
Trigger parameterized build on other projects
Build Triggers
Projects to build
Job B
Trigger when build is
Stable
Trigger build without parameters
Add Parameters
Add trigger...
Add post-build action
Save
Apply
REST API
Jenkins 2.549


LibreWolf
Mon Feb 09 22:38
http://localhost:8090/job/Job B/configure
Import bookmarks...
Jenkins / Job B / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
General
Enabled
Description
This is a Test project
Plain text Preview
Discard old builds
GitHub project
This build requires lockable resources
This project is parameterized
Throttle builds
Execute concurrent builds if necessary
Advanced
Source Code Management
Connect and manage your code repository to automatically pull the latest code for your builds.
None
Git
Subversion
Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Save
Apply


LibreWolf
Mon Feb 09 22:39
http://localhost:8090/job/Job B/configure
Import bookmarks...
Jenkins / Job B / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute shell
Command
See the list of available environment variables
echo "This is a TEST project item"
Advanced
Add build step
Post-build Actions
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
Trigger parameterized build on other projects
Build Triggers
Projects to build
Job C
Trigger when build is
Stable
Save
Apply


LibreWolf
Mon Feb 09 22:39
http://localhost:8090/job/Job B/configure
Import bookmarks...
Jenkins / Job B / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Advanced
Add build step
Post-build Actions
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
Trigger parameterized build on other projects
Build Triggers
Projects to build
Job C
Trigger when build is
Stable
Trigger build without parameters
Add Parameters
Add trigger...
Add post-build action
Save
Apply
REST API
Jenkins 2.549

LibreWolf Mon Feb 09 22:40

http://localhost:8090/job/Job C/configure

Import bookmarks...

Jenkins / Job C / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled

Description

This is a Deploy project

Plain text Preview

- [ ] Discard old builds ?
- [ ] GitHub project
- [ ] This build requires lockable resources
- [ ] This project is parameterized ?
- [ ] Throttle builds ?
- [ ] Execute concurrent builds if necessary ?

Advanced

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

- (•) None
- ( ) Git ?
- ( ) Subversion ?

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

Save Apply

LibreWolf Mon Feb 09 22:41

http://localhost:8090/job/Job C/configure

Import bookmarks...

Jenkins / Job C / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

- [ ] SSH Agent
- [ ] Terminate a build if it's stuck
- [ ] With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See the list of available environment variables

```
echo "the project is deployed"
```

Advanced

+ Add build step

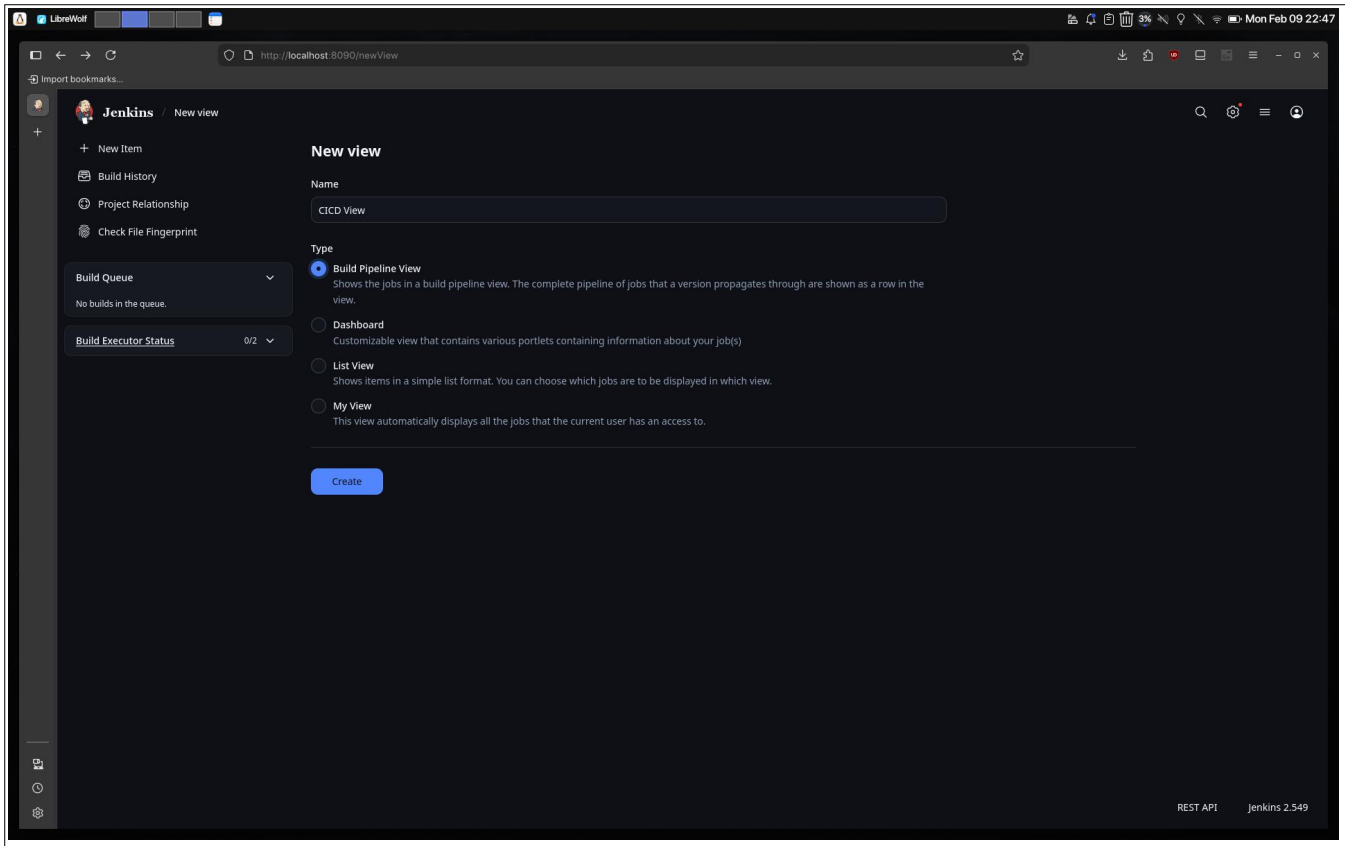
Post-build Actions

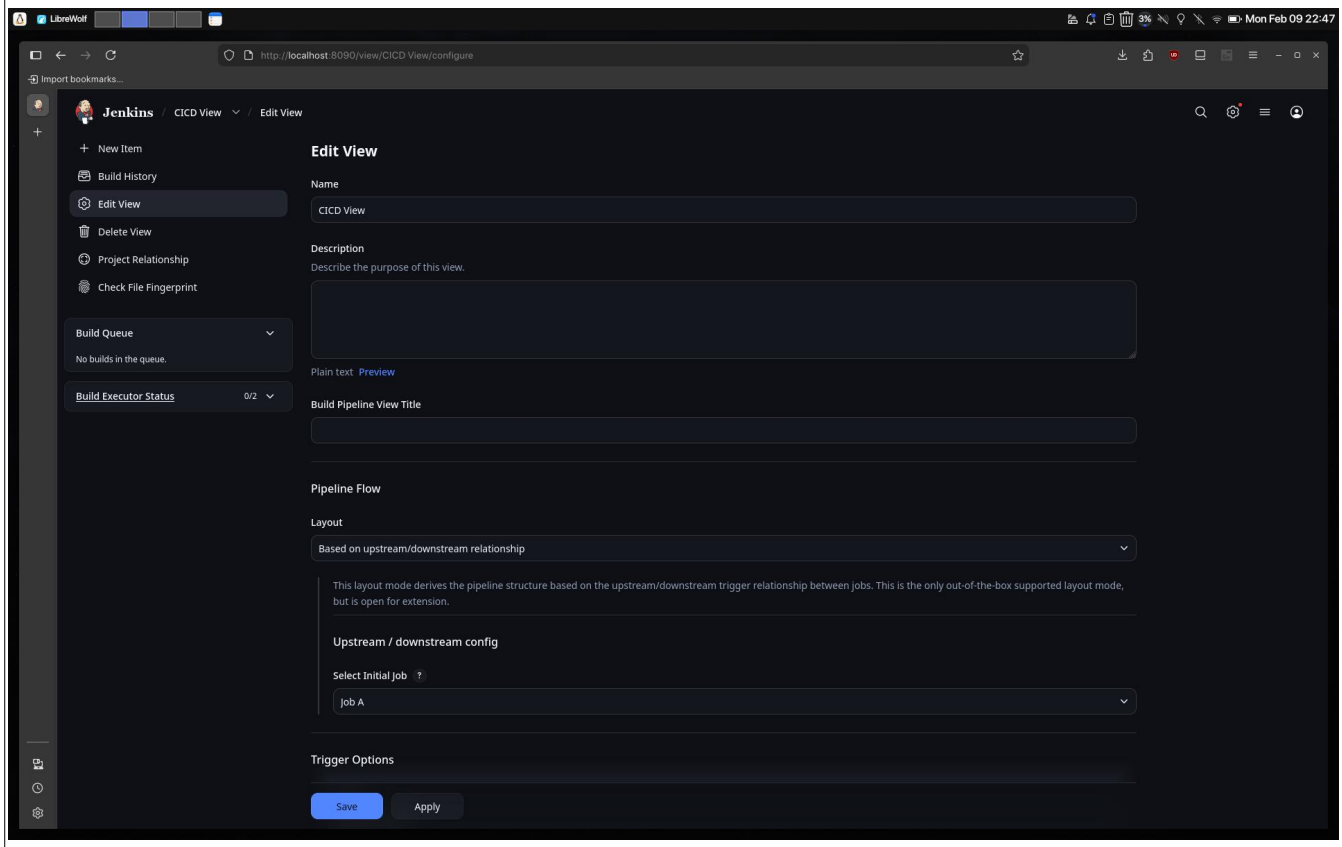
Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

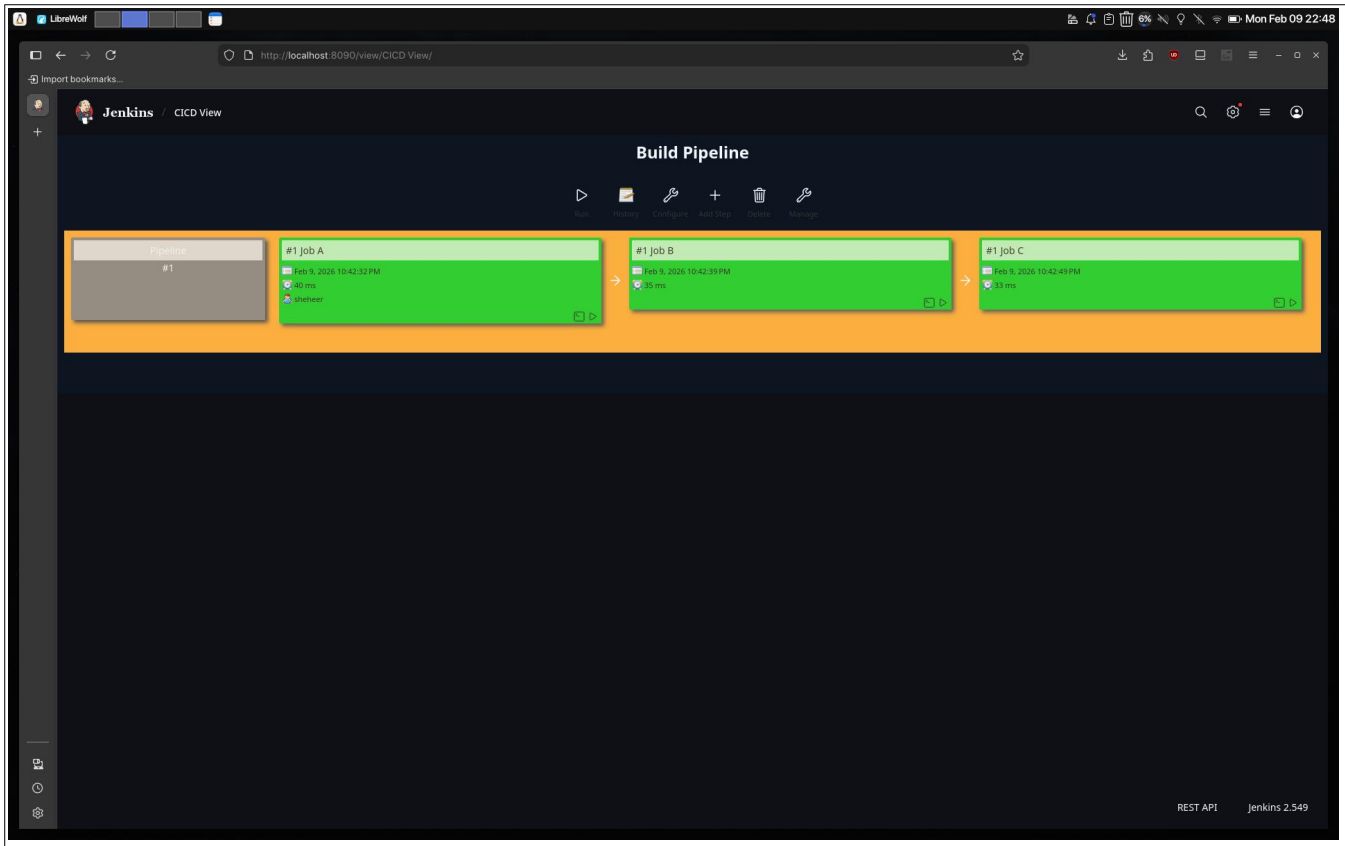
+ Add post-build action

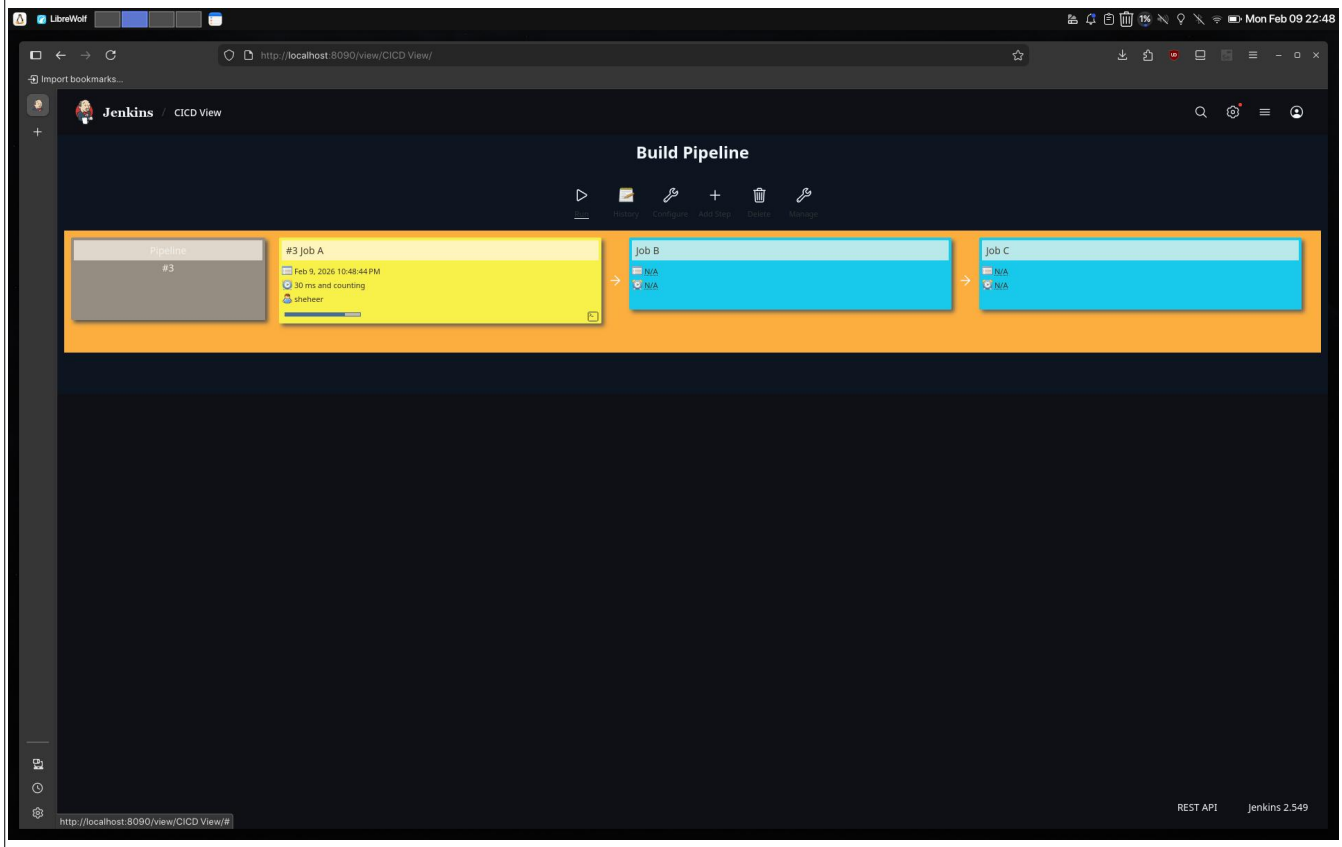
Save Apply

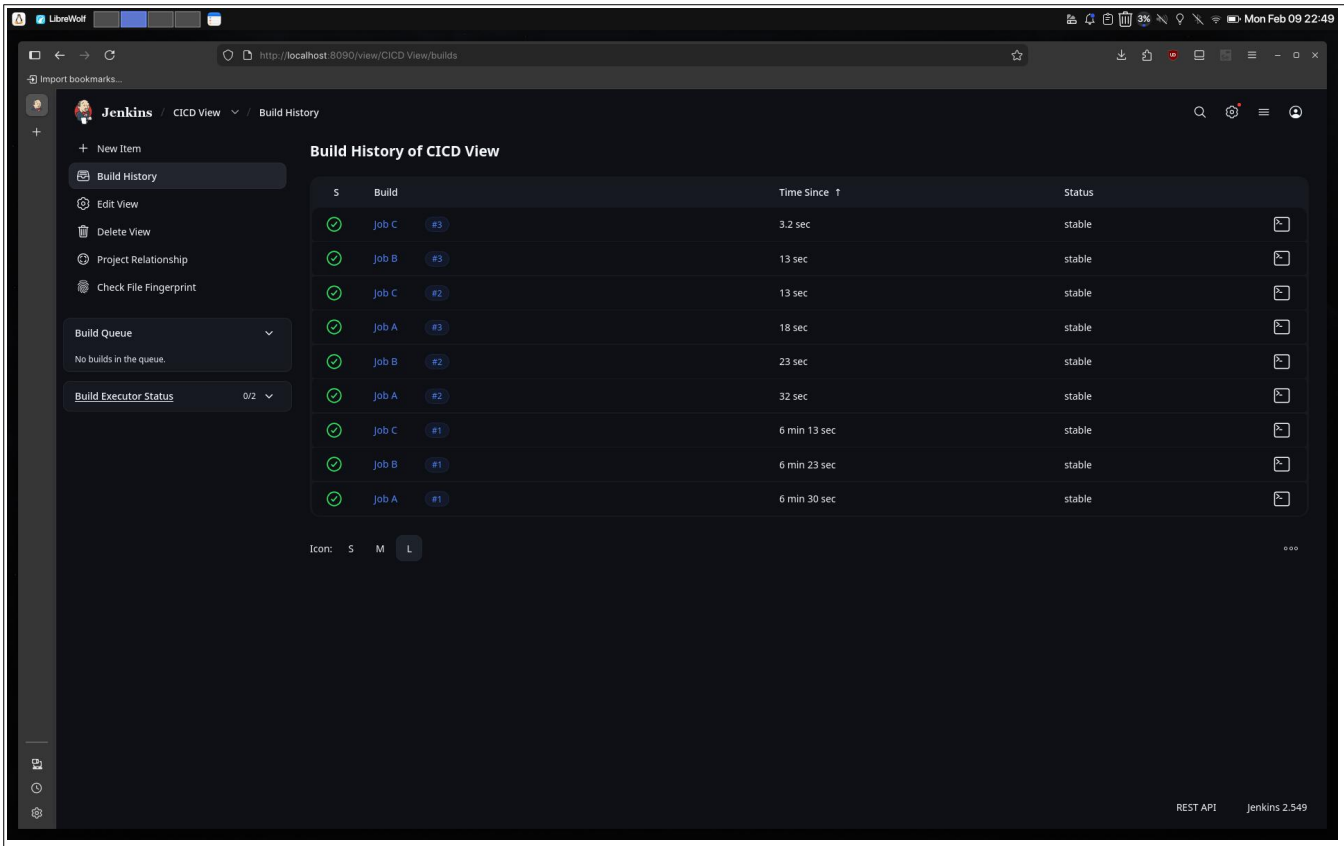
REST API Jenkins 2.549


LibreWolf
Mon Feb 09 22:47
http://localhost:8090/newView
Import bookmarks...
Jenkins / New view
New Item
Build History
Project Relationship
Check File Fingerprint
Build Queue
No builds in the queue.
Build Executor Status 0/2
New view
Name
CICD View
Type
Build Pipeline View
Shows the jobs in a build pipeline view. The complete pipeline of jobs that a version propagates through are shown as a row in the view.
Dashboard
Customizable view that contains various portlets containing information about your job(s)
List View
Shows items in a simple list format. You can choose which jobs are to be displayed in which view.
My View
This view automatically displays all the jobs that the current user has an access to.
Create
REST API
Jenkins 2.549


LibreWolf
Mon Feb 09 22:47
http://localhost:8090/view/CICD View/configure
Import bookmarks...
Jenkins / CICD View / Edit View
New Item
Build History
Edit View
Delete View
Project Relationship
Check File Fingerprint
Build Queue
No builds in the queue.
Build Executor Status 0/2
Edit View
Name
CICD View
Description
Describe the purpose of this view.
Plain text Preview
Build Pipeline View Title
Pipeline Flow
Layout
Based on upstream/downstream relationship
This layout mode derives the pipeline structure based on the upstream/downstream trigger relationship between jobs. This is the only out-of-the-box supported layout mode, but is open for extension.
Upstream / downstream config
Select Initial Job
Job A
Trigger Options
Save
Apply

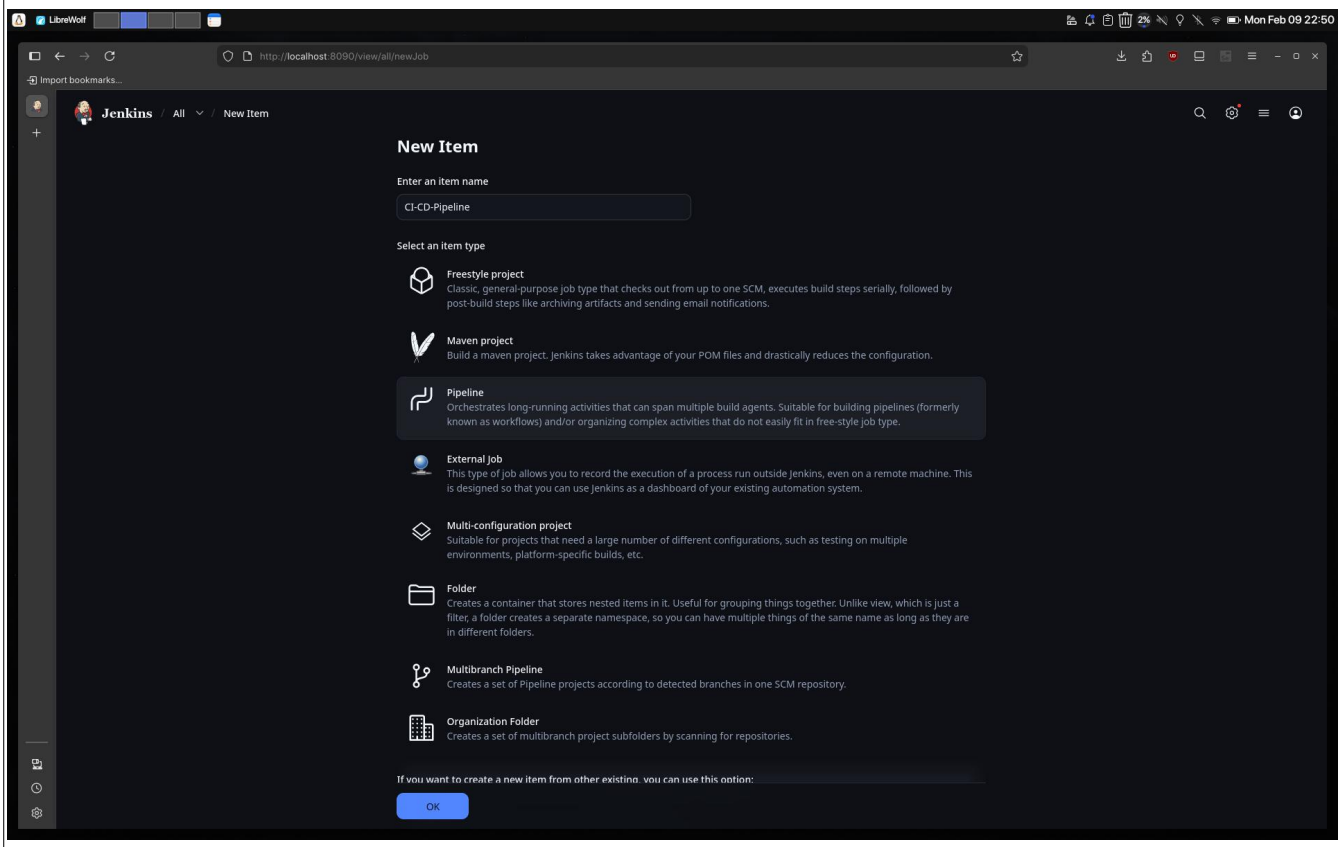

LibreWolf
Mon Feb 09 22:48
http://localhost:8090/view/CICD View/
Import bookmarks...
Jenkins / CICD View
Build Pipeline
Run
History
Configure
Add Step
Delete
Manage
Pipeline
#1
#1 Job A
Feb 9, 2026 10:42:32 PM
40 ms
sheheer
#1 Job B
Feb 9, 2026 10:42:39 PM
35 ms
#1 Job C
Feb 9, 2026 10:42:49 PM
33 ms
REST API
Jenkins 2.549

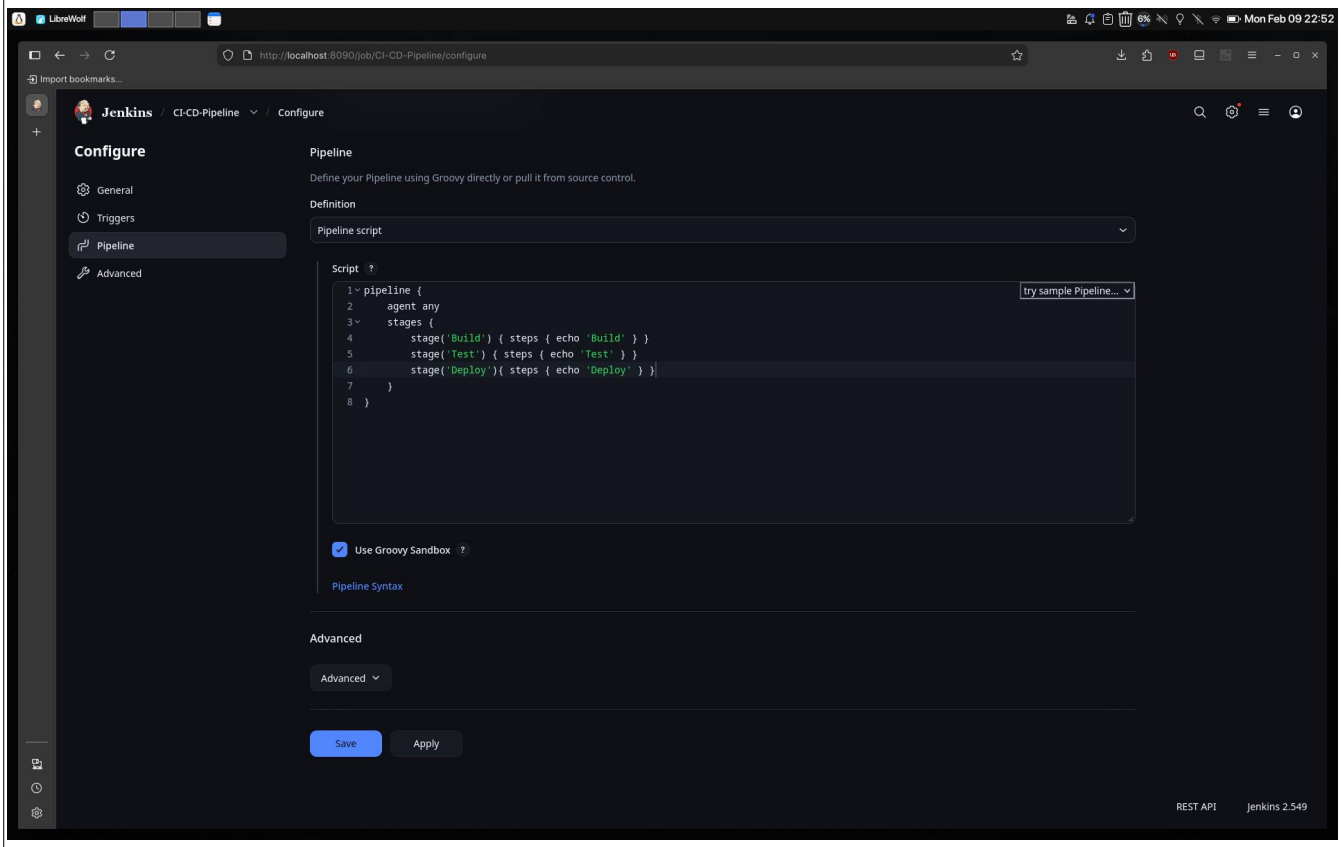

LibreWolf
Mon Feb 09 22:48
http://localhost:8090/view/CICD View/
Import bookmarks...
Jenkins / CICD View
Build Pipeline
Run
History
Configure
Add Step
Delete
Manage
Pipeline
#3
#3 Job A
Feb 9, 2026 10:48:44 PM
30 ms and counting
sheheer
Job B
N/A
N/A
Job C
N/A
N/A
REST API
Jenkins 2.549
http://localhost:8090/view/CICD View/#

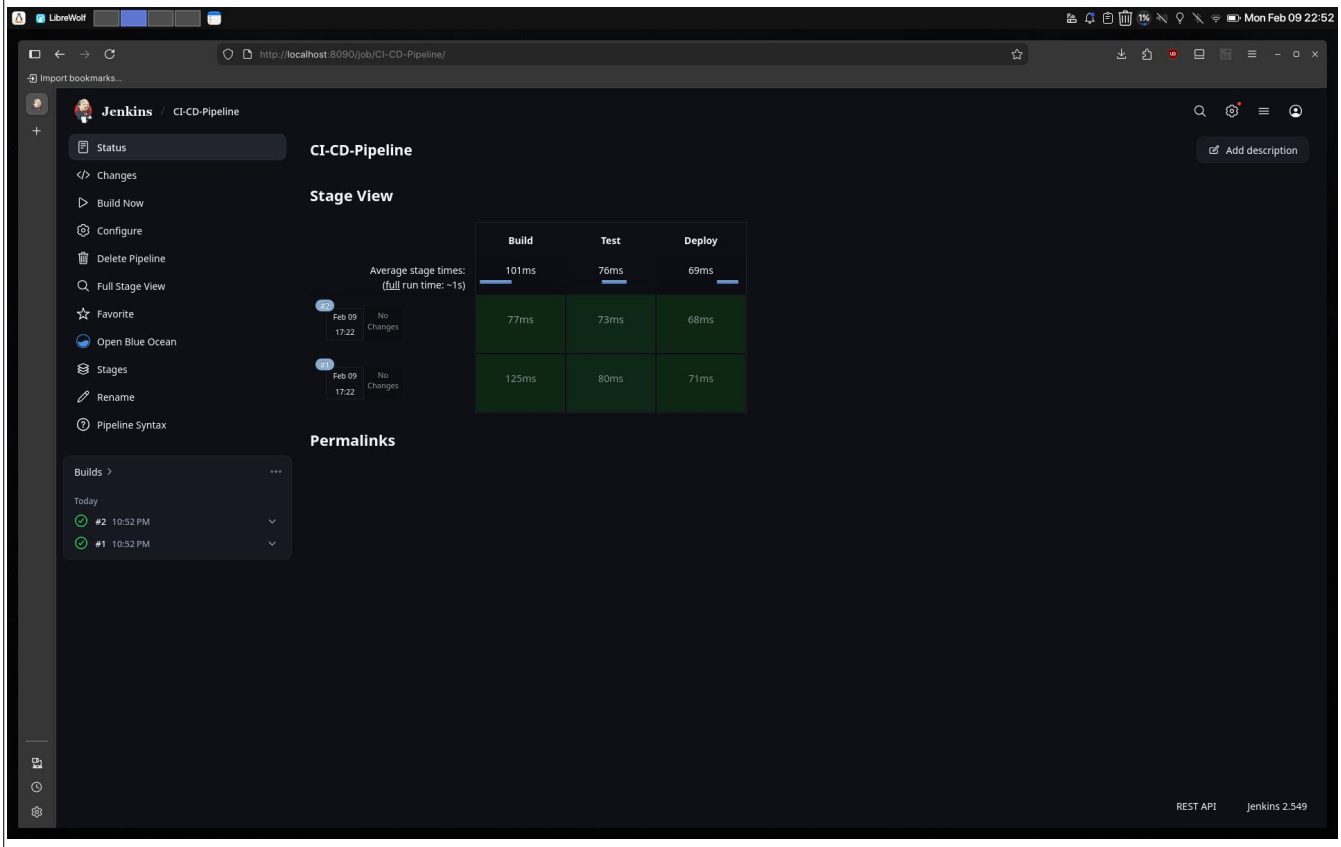

LibreWolf
Mon Feb 09 22:49
http://localhost:8090/view/CICD View/builds
Import bookmarks...
Jenkins / CICD View / Build History
New Item
Build History
Edit View
Delete View
Project Relationship
Check File Fingerprint
Build Queue
No builds in the queue.
Build Executor Status 0/2
Build History of CICD View
S Build Time Since Status
Job C #3 3.2 sec stable
Job B #3 13 sec stable
Job C #2 13 sec stable
Job A #3 18 sec stable
Job B #2 23 sec stable
Job A #2 32 sec stable
Job C #1 6 min 13 sec stable
Job B #1 6 min 23 sec stable
Job A #1 6 min 30 sec stable
Icon: S M L
REST API
Jenkins 2.549

4. Use Jenkins Pipeline & Jenkinsfile.

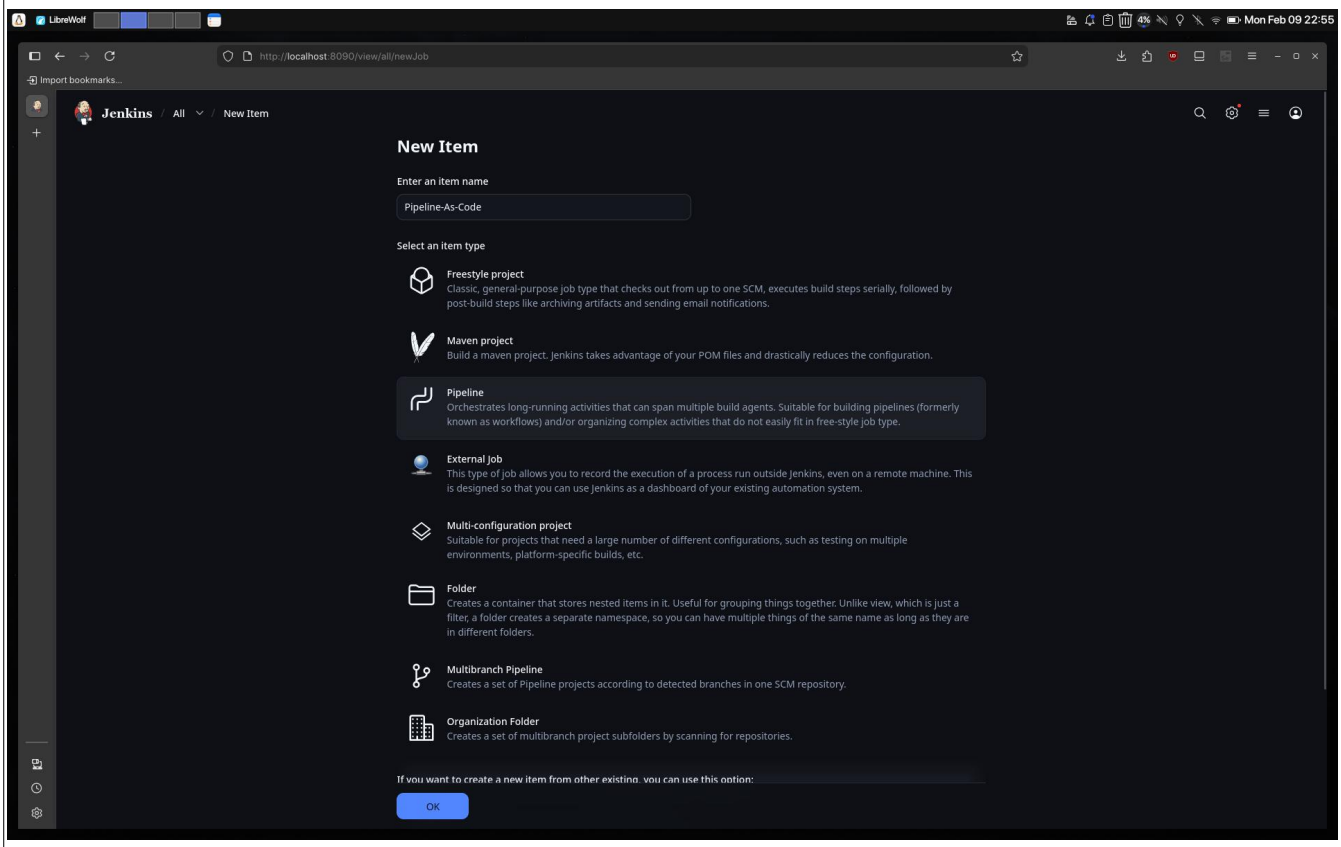
Jenkins Pipeline is a way to define an entire CI/CD workflow as code using a Jenkinsfile, making the build, test, and deploy process automated and repeatable.
Scripted Pipeline is more flexible but complex and written in Groovy, while Declarative Pipeline is simpler, structured, and recommended for most use cases.

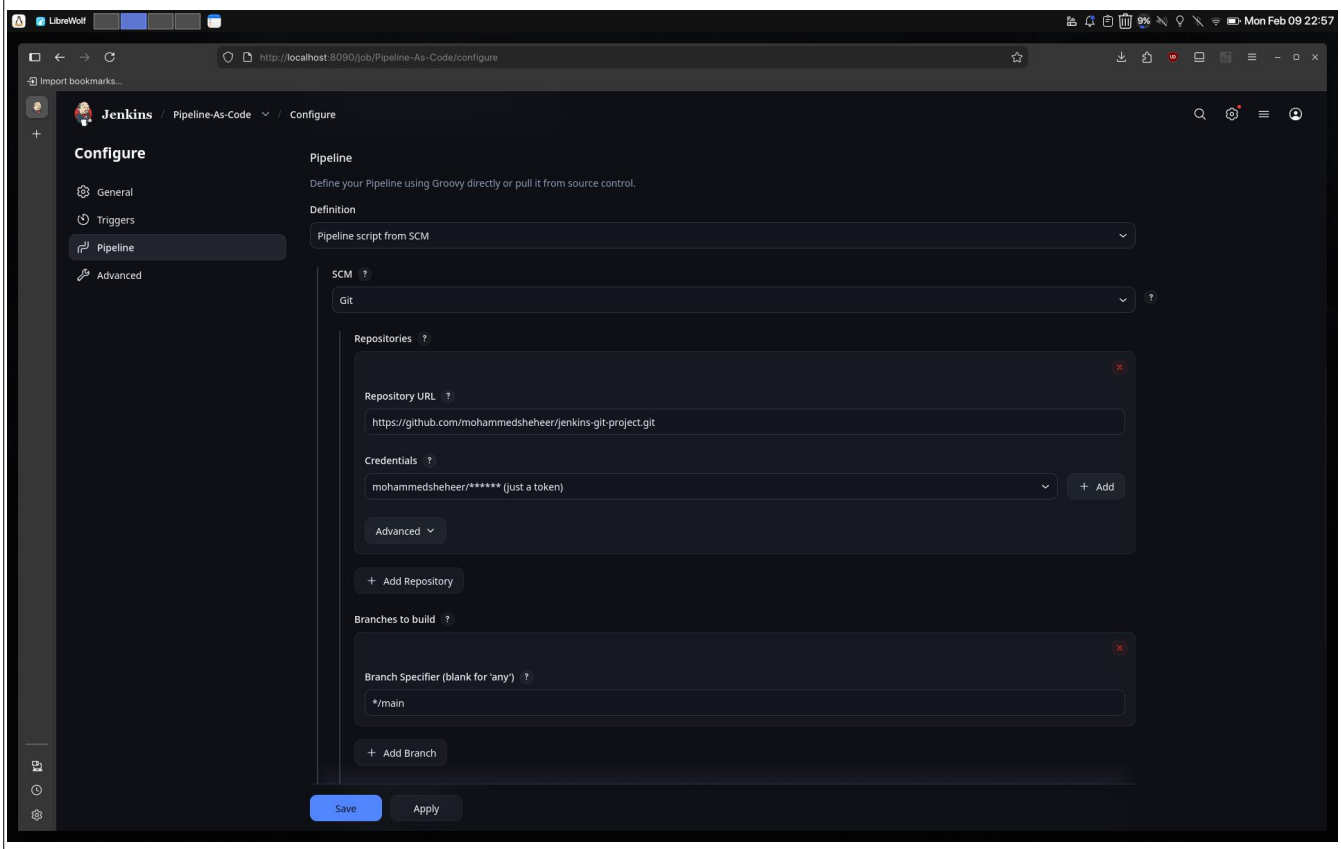


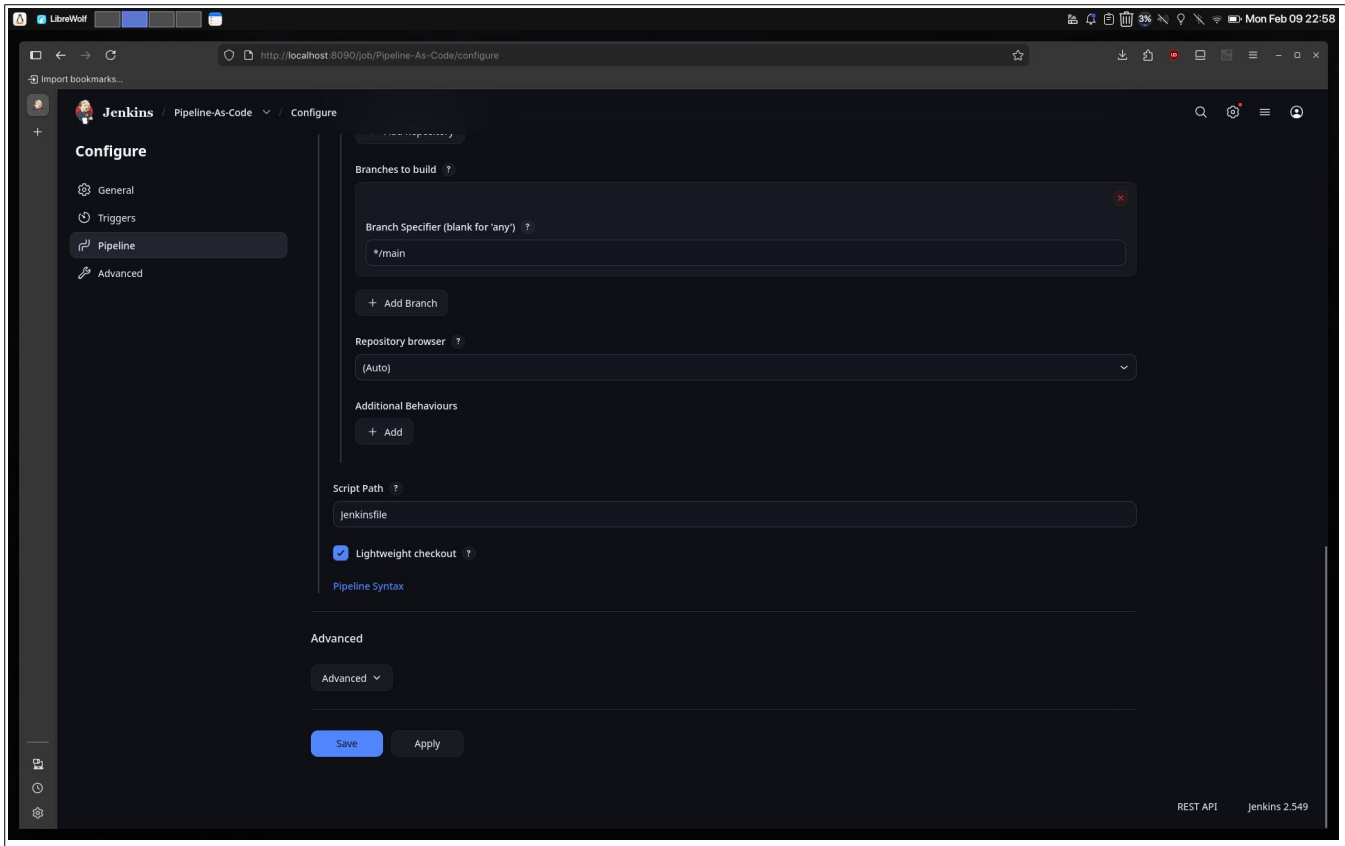


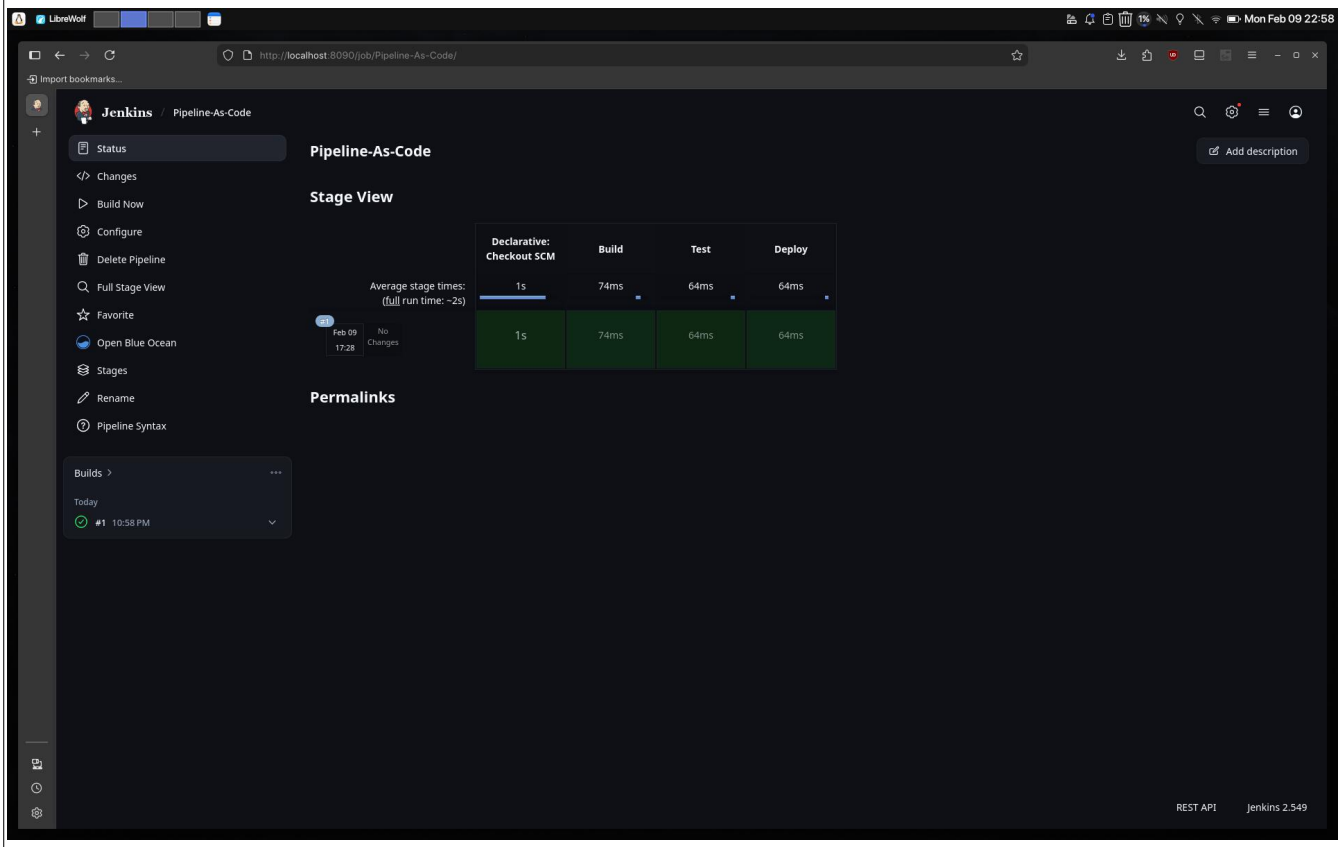

LibreWolf
Mon Feb 09 22:52
http://localhost:8090/job/CI-CD-Pipeline/
Import bookmarks...
Jenkins / CI-CD-Pipeline
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Favorite
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Builds
Today
#2 10:52 PM
#1 10:52 PM
CI-CD-Pipeline
Add description
Stage View
Build
Test
Deploy
Average stage times:
(full run time: ~1s)
101ms
76ms
69ms
#2
Feb 09
17:22
No Changes
77ms
73ms
68ms
#1
Feb 09
17:22
No Changes
125ms
80ms
71ms
Permalinks
REST API
Jenkins 2.549

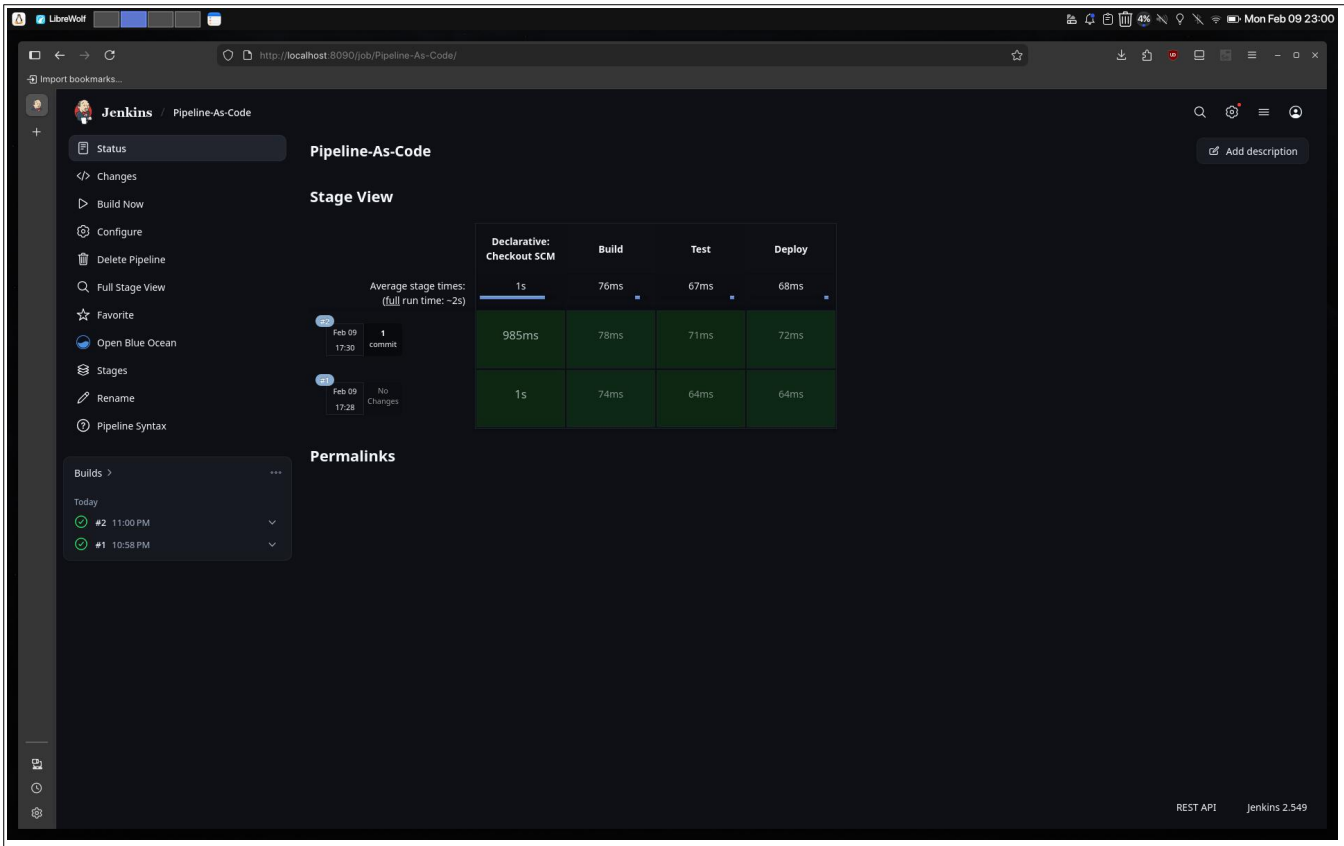
5. Pipeline as code from repository.






LibreWolf
Mon Feb 09 22:58
http://localhost:8090/job/Pipeline-As-Code/configure
Import bookmarks...
Jenkins / Pipeline-As-Code / Configure
Configure
General
Triggers
Pipeline
Advanced
Branches to build
Branch Specifier (blank for 'any')
*/main
Add Branch
Repository browser
(Auto)
Additional Behaviours
Add
Script Path
Jenkinsfile
Lightweight checkout
Pipeline Syntax
Advanced
Advanced
Save
Apply
REST API
Jenkins 2.549


LibreWolf
Mon Feb 09 22:58
http://localhost:8090/job/Pipeline-As-Code/
Import bookmarks...
Jenkins / Pipeline-As-Code
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Favorite
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Builds
Today
#1 10:58 PM
Pipeline-As-Code
Add description
Stage View
Declarative: Checkout SCM
Build
Test
Deploy
Average stage times:
(full run time: ~2s)
1s
74ms
64ms
64ms
#1
Feb 09
17:28
No Changes
1s
74ms
64ms
64ms
Permalinks
REST API
Jenkins 2.549


LibreWolf
Mon Feb 09 23:00
http://localhost:8090/job/Pipeline-As-Code/
Import bookmarks...
Jenkins / Pipeline-As-Code
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Favorite
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Builds
Today
#2 11:00 PM
#1 10:58 PM
Pipeline-As-Code
Add description
Stage View
Declarative: Checkout SCM
Build
Test
Deploy
Average stage times:
(full run time: ~2s)
1s
76ms
67ms
68ms
#2
Feb 09
17:30
1
commit
985ms
78ms
71ms
72ms
#1
Feb 09
17:28
No Changes
1s
74ms
64ms
64ms
Permalinks
REST API
Jenkins 2.549